Less to Lose:

How Shorter and More Flexible Degrees Shape University Choice and Earnings

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Abstract

Making university degrees shorter or adding intermediate credentials can be effective tools to increase university participation. However, a short degree provides less training and an intermediate credential may induce students who would have otherwise completed the full degree to stop early. Understanding how students navigate these trade-offs is important for the design higher-education systems. I study this using the Bologna Process as a large-scale natural experiment. Using a regression discontinuity across birth cohorts and linked employer–employee data from Portugal, I find that shorter degrees and intermediate credentials increase university attainment by 8%, time spent at university by 0.23 years, and hourly wages by 2.1%. More strikingly, I find that adding an intermediate credential inside a five-year degree increases the share of people completing the full five-years. I explain this finding with a dynamic discrete-choice model: the flexibility of intermediate credentials attracts students who are less confident, some of whom discover that they are good and complete the full degree. Counterfactuals show that whether an intermediate credential raises full degree completion depends on when it is awarded and its returns relative to the full degree.

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1 Introduction

Higher-education systems differ substantially, both between and within countries. An undergraduate degree in economics takes three years in England, four in the United States and four or five in many Latin American countries. Degree structures also differ within countries. In several European countries, economics students earn a bachelor's degree after three years and can then choose whether to continue to a master's, whereas law students enrol in a single five-year programme and receive their first degree only at the end. This paper evaluates the trade-offs between these designs by studying how students' education choices and careers respond when degrees become shorter and more flexible.

Length matters because university is costly: students pay fees and forgo earnings for every year they study. A shorter first degree therefore lowers the cost of obtaining a qualification and can attract students who would otherwise not enrol, but it may also send graduates into the labour market with less training. Flexibility matters because university is also uncertain: students enrol unsure whether it will suit them and learn about their fit only while studying. Because much of the return depends on obtaining a qualification (Hungerford and Solon, 1987; Jaeger and Page, 1996), a student who leaves without one may recover little of what she spent. Being able to leave with a bachelor's after three years reduces this loss. This can draw in new students, some of whom turn out to be well matched and complete the long programme, but it can also lead students who would otherwise have completed it to stop early. Because fields within a system are often structured differently, length and flexibility may also affect what students choose to study. With these forces pulling in different directions, the overall effects of shorter and more flexible degrees on education and earnings are not obvious ex ante.

The Bologna Process provides a large-scale natural experiment for studying these trade-offs. In 1999, 29 European countries agreed to move from their higher-education systems towards a common two-cycle structure: a bachelor's degree followed by a master's (European Ministers of Education, 1999). Because fields entered the reform with different structures, Bologna changed them in different ways. In fields that already had a first degree followed by a master's, the reform typically shortened the first degree, so the main change was its length. In fields with a single long degree, it split the degree into a bachelor's and a the master's, so the main change was flexibility: students gained an intermediate credential and could decide to

stop or continue. In many countries, fields such as medicine were left unchanged.

The reform sparked a heated public debate. Advocates argued that shorter and more flexible degrees would widen access to university. Critics worried that graduates would receive less training, that employers would not value the new bachelor's, and that students would stop at the bachelor's instead of completing the full degree (Cappellari and Lucifora, 2009). Yet causal evidence on the reform's effects beyond enrolment remains scarce (Kroher et al., 2021). The debate therefore remains open and is increasingly relevant beyond Europe, with Argentina, Chile and Peru shortening degree programmes, introducing intermediate credentials or considering such reforms.[1]

This paper provides causal evidence to inform such decisions by exploiting the rapid rollout of the reform in Portugal and Italy. Using a regression discontinuity design, I compare birth cohorts that reached university age just before and just after the reform. I then use differences in fields' pre-reform structures to distinguish responses to a shorter first degree from responses to a new intermediate credential. By analysing cohorts as a whole rather than conditioning on enrolment or graduation, I allow the reform to affect whether and what individuals study, where they work and what they earn. Portugal is the main setting because its unusually rich linked employer–employee dataset, *Quadros de Pessoal*, covers private-sector employees and records their education, hourly wages, occupations, sectors and employers. Because the reform may also affect selection into private-sector employment, I assess the robustness of the main findings using the nationally representative Portuguese Labour Force Survey. Italian Labour Force Survey data provide an external validity assessment under a different reform date and macroeconomic environment.

The reduced-form analysis yields three main findings. First, the reform raised both educational attainment and wages, especially among women. In Portugal, university completion rose by 2.4 percentage points, or 8 percent of the pre-reform mean, and average years of university increased by 0.23. Hourly wages at ages 28 to 30 were 2.1 percent higher; the ratio of the wage and schooling effects is about 9 percent per additional year of university. Italian data show similar gains, concentrated among women. Second, the reform changed what individuals studied. Measured over the whole cohort, the probability of studying a field that gained an

[1] In Argentina, Ministerial Resolution 556/2025 introduced a national credit framework under which new undergraduate programmes must comprise at least four years and 240 credits, scheduled to apply from 2027; universities including UBA have also shortened some programmes and introduced intermediate credentials. In Peru, a bill presented in 2024 proposed a technical credential after six semesters of university. In 2026, Chile's Ministry of Education and Council of Rectors began discussing more flexible pathways and the duration of undergraduate degrees.

intermediate credential rose by 27 percent relative to its pre-reform mean, while the probability of studying a field whose first degree became shorter rose by 15 percent. The probability of studying a field left unchanged by the reform fell by 31 percent. These shifts carried over into jobs: post-reform cohorts were more likely to work in sectors and occupations related to the fields that expanded, suggesting that the additional graduates found work related to their studies. Third, contrary to what critics predicted, introducing the option to stop after three years increased full-programme completion. In fields that gained an intermediate credential, the share of the cohort completing all five years rose by 0.7 to 1.1 percentage points, depending on the measure.

To understand these responses, I build a dynamic discrete choice model in which students learn about their fit with university as they progress. An intermediate credential gives students who leave early less to lose, drawing in those who are less certain that they will succeed. Some of these new entrants discover that they are well matched and complete the full programme, so five-year completion can rise even though other students now stop at the bachelor's. I estimate the model using the reform effects and validate it against two untargeted outcomes. The model predicts that the bachelor's wage is 87.4 percent of the five-year-degree wage and that cohort wages rise by 1.8 percent, compared with 86.8 and 2.1 percent in the data.

Counterfactuals show that an intermediate credential raises full-degree completion only if it arrives early enough to insure students but pays sufficiently less than the full degree to preserve the incentive to continue. Moving the intermediate credential from year three to year four would insure too few students to have much effect, raising the probability of studying these fields by at most 0.4 percentage points, compared with 2.3 percentage points under the observed reform. For countries now introducing intermediate credentials, the model therefore shows that whether they raise full-degree completion depends on when they are awarded and their returns compared to the full degree.

This paper makes three contributions. First, it contributes to the study of how to design higher-education systems. Existing work examines when students choose a field (Malamud, 2010; Bordon and Fu, 2015), how they are admitted and assigned to programmes (Hastings et al., 2013; Kirkeboen et al., 2016), which tiers and sectors the system offers (Goodman et al., 2017; Mountjoy, 2022), and how it is priced (Epple et al., 2006; Fu, 2014).[2] Much less is known about degree length and the availability of intermediate credentials.

[2]For reviews, see Altonji et al. (2016) and Lovenheim and Smith (2023).

The few causal studies of degree length compare graduates and therefore hold enrolment fixed (Arteaga, 2018; Gong and Pan, 2023). The closest evidence on intermediate credentials comes from secondary school: when German states gave academic-track students a qualification after grade 9, more students completed the full track and entered university (Obergruber and Zierow, 2020). I study both design margins in higher education, exploiting a reform that shortened the first degree in some fields and introduced an intermediate credential in others. The relative increase in studying affected fields was larger where the reform introduced an intermediate credential than where it only shortened the first degree.

Second, the paper contributes to the study of the option value of education using a policy that changed the option itself. A large literature models education as experimentation, in which the options to continue and to leave have value (Manski, 1989; Stinebrickner and Stinebrickner, 2012; Stange, 2012; Trachter, 2015; Arcidiacono et al., 2025). The structural models in this literature are generally estimated using observational data and quantify the value of options that already exist; the closest exception validates a model of sequential schooling choices against a compulsory-schooling reform (Bhuller et al., 2024). Bologna instead created an option – the ability to leave with a credential while retaining the opportunity to complete the long programme – and, to my knowledge, this is the first paper to estimate such a model using a reform that changes students' options. This option also differs from the stepping stone studied most extensively. Community colleges offer a cheaper first step at a separate institution, and research asks whether they draw new students into higher education or divert them from a four-year degree (Rouse, 1995; Mountjoy, 2022). Bologna placed the stepping stone inside the same university programme, where, on net, it attracted new students and increased completion of the full degree.

Finally, the paper provides, to my knowledge, the first causal evaluation of the Bologna Process that estimates effects at the level of birth cohorts and follows them into the labour market. Existing research has focused mainly on enrolment and on dropout or on-time graduation among those who enrol (Cappellari and Lucifora, 2009; Cardoso et al., 2008; Di Pietro and Cutillo, 2008; Horstschräer and Sprietsma, 2015; Bondonio and Berton, 2018). Evidence on fields of study, careers and earnings is limited and mainly descriptive (see Kroher et al., 2021, for a review), and it generally conditions on graduation (Bosio and Leonardi, 2010; Glauser et al., 2019; Suleman and Figueiredo, 2020), thereby excluding the marginal students whom the

reform drew into university. The linked employer–employee data also extend the evidence beyond earnings. Unlike the unemployment-insurance records commonly used in the community-college literature (Jepsen et al., 2014; Stevens et al., 2019; Mountjoy, 2022), my data records hourly wages, occupations and employers. I can therefore distinguish higher pay from longer hours, examine whether the additional graduates enter jobs related to their fields, and, building on Portugal et al. (2024), measure the role of sorting into higher-paying firms.

The rest of the paper is structured as follows. Section 2 describes the institutional setting. Section 3 presents the data sources and explains how the samples are built. Section 4 sets out the empirical strategy. Section 5 presents the reduced form results, and Section 6 presents the model. Section 7 concludes.

2 Institutional Setting

In 1999, 29 European countries signed the Bologna Declaration, with the goal of creating a European Higher Education Area in which degrees would be comparable and students could move easily between countries. To do this, countries agreed on a common structure with two cycles: a bachelor's degree of at least three years followed by a master's degree on top. Before the reform, degree structures differed both across and within countries (Cardoso et al., 2008). Reaching the agreed structure therefore required different changes depending on the starting point of each field in each country. In some fields the first degree became shorter, in others students gained an intermediate credential partway through a longer degree, and in others nothing changed. Even within a single country, the reform was therefore several reforms at once, which lets me study how students respond to each type of change.

Restructuring by field. Portugal illustrates this well. Before the reform, all university programmes led to a *licenciatura*, but its length and what followed it depended on the field. In fields such as economics, management, the sciences, the social sciences and the humanities, students completed a four-year *licenciatura* and could then enrol in a two-year master's. The reform shortened the *licenciatura* to three years, so the 4+2 structure became 3+2. Programmes had to fit four years of study into three, which they did by dropping more advanced courses and compressing others. Students therefore obtained their first degree a year earlier,

but with less coursework than before. In fields such as engineering, computing and architecture, students instead enrolled in a single five-year *licenciatura*. These programmes became five-year integrated master's degrees. The curriculum stayed largely the same, but it was split into two parts: the first three years led to a *licenciatura* and the last two to a master's. Before the reform, a student who left these programmes after three years had no degree; after the reform, they left with one. The law allowed integrated master's degrees only where a profession requires longer training, either under EU rules or by established European practice. Finally, a third group of fields kept their pre-reform structure: nursing remained a four-year degree and medicine a six-year one. In short, the first type of change made a degree cheaper to obtain, while the second made it less costly to leave early. Figure 1 summarises the three groups.

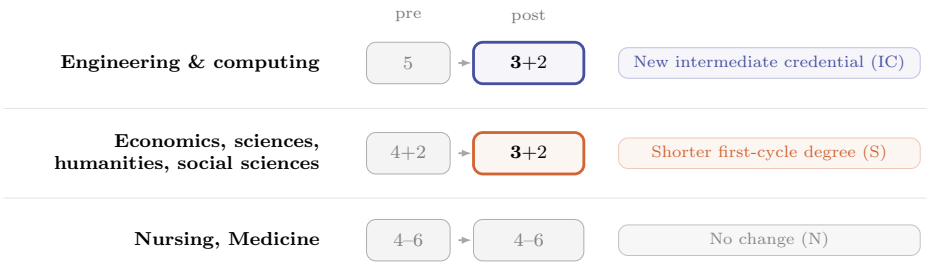


Figure 1: Years to the first university credential in Portugal before and after the reform, by field

Notes: Statutory years to the first degree. (+2): pre-existing master's. Blue: new intermediate credential; orange: shorter first-cycle degree; grey: no change.

Comparison with Italy. How the reform played out in each country depended on its pre-reform structure, and Italy provides a useful contrast. Unlike Portugal, Italy had no master's tier before the reform, and the old four or five-year *laurea* was treated as equivalent to the new master's. As a result, economics, the sciences, the humanities and the social sciences gained an intermediate credential, like engineering, rather than a shorter first degree. The key difference across fields in Italy is therefore whether a field went from 4 to 3+2 or 5 to 3+2. Appendix B describes the Italian reform in more detail.

Timeline. Both roll-outs were fast (Figure 2). In Portugal, Decree-Law 74/2006 let programmes convert from 2006/07 and required them to by 2008/09; 43 percent converted in the first year and about 90 percent by 2007/08 (Cardoso et al., 2008), so the 2007 applicants are the first cohort to face an almost fully reformed system. In Italy, universities could adopt the reform from 2000/01, but most did so in 2001/02 (Cappellari

and Lucifora, 2009; Bondonio and Berton, 2018).

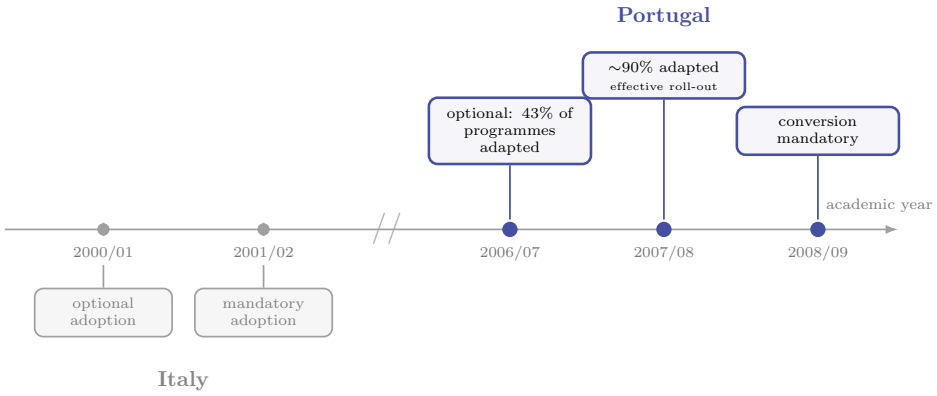


Figure 2: Implementation of the reform in Portugal and Italy

Notes: Sources: Decree-Law 74/2006; Ministerial Decree 509/1999.

Enrolment: demand and supply Enrolment rose after the reform in both countries, and this increase was driven by demand rather than by the supply of places. In Portugal each institution proposes the number of places per programme, and candidates are allocated to them through a national contest. In the first round of the contest, places rose by 5 percent in 2007, while the number of candidates rose by 27 percent (Direção-Geral do Ensino Superior, 2010). The extra students filled places that already existed, since in 2006 one place in four had gone unfilled. In Italy most programmes had no cap on entry, so enrolment was not rationed by places at all. First-time entrants rose by 12 percent in 2001/02 and stayed about 18 percent above their 1998–2000 level (Ministero dell'Università e della Ricerca, 2024).

3 Data

3.1 Data sources

I combine one administrative data set and three surveys. *Quadros de Pessoal* is the backbone of the Portuguese analysis: it follows every private-sector employee over time and records education, field of study, occupation, job title, pay and employer for the full cohorts on both sides of the reform. Two Portuguese surveys fill the gaps it leaves. The nationally representative Labour Force Survey covers public-sector workers, the self-employed and those out of work, and the Education and Training Survey records dropout rates.

The Italian Labour Force Survey supplies the Italian analysis, which tests whether the Portuguese results hold in a country that reformed six years earlier.

Quadros de Pessoal. *Quadros de Pessoal* (QP) is a linked employer–employee data set collected every year by the Portuguese Ministry of Labour from all private-sector establishments with at least one employee. Reporting is compulsory, and each establishment reports its workforce as of October. QP therefore covers the full population of workers and firms in manufacturing and private services; it excludes civil servants, the self-employed and domestic workers. Public hospitals are nonetheless largely covered, because most of them are public enterprises (*Entidades Públicas Empresariais*, EPE), which report to QP. For each worker I observe sex, birth date, the highest completed level of education and its field, a four-digit occupation code, the job title within the applicable collective agreement (*categoria profissional*), the worker's hierarchical level within the firm (*nível de qualificação*), normal monthly hours, and monthly base pay, regular supplements and overtime pay. Workers and firms carry anonymised identifiers, so I can follow a worker across employers and years. I use the waves from 2000 to 2023; no worker file exists for 2001. The QP education variable records well whether a worker holds a university degree, but not which degree. Holding any degree is therefore the main QP education outcome. I proxy years of university with the age at which a worker is first observed with their highest degree, minus 18, for workers who hold a degree, and zero for everyone else. Because this also counts any delay before the first private-sector job, I validate it against the Portuguese Labour Force Survey, which records the age at which a person left education (Appendix C.1).

Portuguese Labour Force Survey (LFS). The *Inquérito ao Emprego* is the quarterly household survey of Statistics Portugal (INE). It is representative of the resident population, and I use the quarterly files from 1998 to 2024. Unlike QP, it covers people in every employment status, so it measures educational attainment without conditioning on private-sector employment, or on employment at all. It records the highest completed level of education and its broad field, the age at which the respondent left education, employment status and monthly pay. Its education codes are close to those in QP. The LFS can also tell pre-reform from post-reform *licenciaturas*, which matters for measuring five-year degrees. In engineering, the pre-reform *licenciatura* took five years, as the post-reform *mestrado* does. The education code, however, gives every *licenciatura* the same

code, whether it took five years before the reform or three years after it. From 2014 to 2019, INE asks every respondent whose highest degree is a *licenciatura* whether they completed it before or after the reform. I count an engineering degree as a five-year degree if it is a *mestrado* or a doctorate, or a *licenciatura* the respondent reports as pre-reform.

Education and Training Survey (IEFA). The *Inquérito à Educação e Formação de Adultos* is INE's module of the European Adult Education Survey. I use its four waves, 2007, 2011, 2016 and 2022. It is the only Portuguese source that records whether a respondent ever enrolled in higher education and left without a degree. The samples are small—about 3,600 women and 3,500 men in the cohort window—so I use IEFA to study dropout rather than for precise effects on attainment.

Italian Labour Force Survey. The *Rilevazione sulle Forze di Lavoro* is the quarterly household survey of the Italian statistics office (ISTAT). I use the public-use files from 2009, the first year that reports year of birth, to 2020. Like the Portuguese survey, it records the highest completed level of education and its subject, the age at which the respondent left education, employment status, occupation and, for employees, net monthly pay in the main job, top-coded at 3,000 euros. It separates the two-cycle degrees created by the reform—the three-year *laurea* and the two-year *laurea specialistica*—from old-regime and single-cycle degrees, so it measures the length of a degree directly.

3.2 Samples

The design compares people who reached the age of university entry just before and just after the reform, so every sample is built around entry cohorts. I assign each person to the calendar year in which they reached the usual age of entry: 18 in Portugal and 19 in Italy. Data availability then decide at what ages I can observe them. Older ages are better: by then most people who went to university have finished. I therefore measure outcomes at the oldest ages at which the cohorts on both sides of the cutoff can be observed. Because the two reforms are six years apart, these ages differ between the countries.

Portugal. I keep the six cohorts that turned 18 in 2000–2005 and the six that turned 18 in 2007–2012, born 1982–1987 and 1989–1994, and leave out the 2006 cohort, which was only partly exposed (Section 2).

The same twelve cohorts define every Portuguese sample. In *Quadros de Pessoal* the limit on ages is the last wave: the youngest cohort, born in 1994, is 28 in 2022 and 29 in 2023. Unless otherwise stated, I keep one observation per worker. Education is measured at age 28, and the wage is averaged over the years in which the worker is aged 28 to 30, which for the youngest cohort means 28 and 29 only. Wider windows give similar wage effects (Appendix D.5). As in Mountjoy (2022), the reason for measuring outcomes by or just after age 28 is that the share holding a degree has stopped rising by then, as shown in Appendix Figure 13. The resulting sample size is of 1,017,067 workers in QP, 42,278 individuals in LFS and 7,500 respondents in IEFA.

Italy. The Italian reform did not reach every student in the same year: universities in four regions adopted it in 2000/01 and the rest in 2001/02 (Section B). The cutoff therefore depends on where a person studied. I keep Italian citizens in the four cohorts before adoption and the four from adoption onwards: born 1977–1984 in the early-adopting regions and 1978–1985 elsewhere. In Italy the limit on ages is the first year of data. The survey reports year of birth only from 2009, when the oldest cohorts are already 31 or 32, so I use respondents aged 31-35. These ages also suit the Italian setting, where students enter university later. The resulting sample size is of 269,026 observations.

4 Empirical Strategy

Because the reform was rolled out quickly (Section 2), I compare birth cohorts that reached university age just before and just after it. A person in Portugal who turned 18 in 2005 chose whether to go to university under the old system. A person who turned 18 in 2007 made the same choice under the new one.

Let c_i be the calendar year in which person i turns 18, the usual age of application in Portugal. The running variable is $Z_i = c_i - 2007$, and treatment is $D_i = \mathbf{1}[Z_i \geq 0]$: the cohort turning 18 in 2007 is the first to apply to a fully reformed system. Let $Y_i(1)$ and $Y_i(0)$ be the outcomes person i would have with and without the reformed degree structure, and let $\mu_+(z) = \mathbb{E}[Y_i(1) \mid Z_i = z]$ and $\mu_-(z) = \mathbb{E}[Y_i(0) \mid Z_i = z]$. The parameter of interest is the effect of the reform on the first reformed cohort,

$$\tau = \mu_+(0) - \mu_-(0). \tag{1}$$

Everyone with $Z_i \geq 0$ faced the new structure and everyone with $Z_i < 0$ the old one, so the data reveal μ_+ to the right of the cutoff and μ_- to the left. With a continuous running variable, continuity of μ_+ and μ_- at the cutoff would identify τ as the gap between the limits of $\mathbb{E}[Y_i \mid Z_i = z]$ on either side (Hahn et al., 2001). Here Z_i takes only a few values, so $\mu_-(0)$ has to be extrapolated from the pre-reform cohorts. As is common with a discrete running variable (Kolesár and Rothe, 2018), I assume that μ_+ and μ_- are linear in z within the estimation window. The identifying assumption is therefore that, without the reform, the first reformed cohorts would have continued the linear trend of the pre-reform cohorts.

I estimate τ with a linear regression that allows a separate slope on each side of the cutoff:

$$Y_i = \alpha + \tau D_i + \beta_1 Z_i + \beta_2 D_i Z_i + \varepsilon_i, \qquad Z_i \in \mathcal{Z} = \{-7, \ldots, -2\} \cup \{0, \ldots, 5\}. \tag{2}$$

The pre-reform line is $\alpha + \beta_1 Z_i$ and the post-reform line is $\alpha + \tau + (\beta_1 + \beta_2) Z_i$, so τ is the gap between the two at $Z_i = 0$. The window $\mathcal{Z}$ holds six cohorts on each side: those turning 18 in 2000–2005 and in 2007–2012. It leaves out the 2006 cohort ($Z_i = -1$), which applied when 43 percent of programmes had converted. That cohort was only partly exposed, so I drop it rather than assign it to either side, as in a donut design. I estimate equation (2) for the whole sample and separately for women and men.

I follow Mountjoy (2022) and keep one observation per person unless otherwise stated, with each outcome measured at the same ages in every cohort: education at age 28, and wages averaged over ages 28 to 30 (Section 3.2 explains the choice of ages). Because every cohort is observed at the same ages, equation (2) needs no age controls. Outcomes are the education measures and labour-market outcomes defined in Appendix C.1.

Wages raise one further issue. At fixed ages, a person's cohort also fixes the calendar years in which their wage is paid. The pre-reform cohorts are aged 28 to 30 between 2010 and 2017, whereas the post-reform cohorts reach these ages between 2017 and 2023. A jump in wages at the cutoff could therefore reflect the business cycle and not the reform. To remove these calendar-time effects, I measure each wage relative to the wages of workers aged 40 to 45 in the same calendar year. Appendix D.4 defines this wage measure and explains the comparison.

Because treatment varies only across cohorts, I estimate equation (2) on cohort means. I average Y_i within each of the twelve cohorts and fit the equation to these means by ordinary least squares, which gives each cohort equal weight. The cohort is the unit of inference: following Donald and Lang (2007), I use the

conventional standard errors of this regression and a t distribution with eight degrees of freedom, twelve cohorts minus four parameters (Appendix D.2).

The identifying assumption cannot be tested directly, but four pieces of evidence support it. First, people cannot choose the year they were born, and the number of workers per cohort in *Quadros de Pessoal* shows no break at the cutoff (Figure 14, Appendix D.3). Second, I find no evidence that students who turned 18 before 2007 waited to enter the new degrees. If they had, more of them would have worked between leaving secondary school and graduating from university. That share shows no statistically significant change at the cutoff: the estimated jump is 0.1 percentage points (standard error 0.1) on a base of 10.5 percent, and it is also insignificant for women and for men separately (Table 7, Appendix D.3). Waiting would also have raised the age at entry, which barely moved. Among entrants, the mean age at first entry was 18.84 in 2005/06, the only pre-reform year available, and 18.88 on average in 2007/08–2010/11 (Eurostat, 2026). Third, *Quadros de Pessoal* sees only private-sector employees, so a reform that changed who works in the private sector would show up as a change in the sample. I show that key education results replicates in the Labour Force Survey, which covers the whole population (Appendix I), and that the share of people employed does not change at the cutoff. Fourth, the jump is specific to 2007. When I move the cutoff to cohorts where nothing changed, the placebo jumps are mostly insignificant and smaller than the actual one (Appendix G).

For Italy I estimate the same equation. Italian students usually enter university at 19, so c_i is the year a person turns 19. Because the reform reached regions in different years, the cutoff depends on when universities in a person's region of residence adopted it (Section 3.2):

$$Z_i = \begin{cases} c_i - 2000 & \text{if } i \text{ lives in an early-adopting region,} \\ c_i - 2001 & \text{if } i \text{ lives in a late-adopting region.} \end{cases} \tag{3}$$

Treatment is again $D_i = \mathbf{1}[Z_i \geq 0]$. The window is $\mathcal{Z} = \{-4, \ldots, 3\}$, four cohorts on each side, and unlike in Portugal it excludes no cohort around the cutoff because a clear mapping of who was exposed in which year allows for re-centering of the running variable.

5 Results

This section presents the evidence for the reduced-form findings of the introduction. Sections 5.1 to 5.3 cover Portugal. Section 5.1 shows that, on net, shorter and more flexible degrees raised educational attainment and wages, and measures how much of the wage gain comes from higher-paying firms. Section 5.2 shows that students moved towards both kinds of reformed field and away from unchanged ones, and that these shifts carry over to the sectors and occupations where they work. Section 5.3 shows that, in engineering, flexibility raised the number of students completing the full programme. Each of these subsections reports the effects for everyone first and then compares women and men. Section 5.4 repeats these three sets of results on Italian data to test their external validity. Each estimate is the coefficient τ in equation (2), estimated as described in Section 4, and "baseline" is the pre-reform mean of the outcome.

5.1 More graduates, more years of university and higher pay

Figure 3 shows the RD graphs for the four main outcomes of this subsection: completion and dropout in the top row, years of university and the hourly wage in the bottom row.

The share of people completing a university degree increased by 2.4 percentage points with the reform, from a baseline of 31 percent, an 8 percent increase. The Labour Force Survey, which covers the whole population and not only private-sector employees, shows a similar jump (Appendix Table 12), so it does not come from a change in who works in the private sector. At the same time, dropout rose by a similar amount. Only IEFA, a much smaller survey, records this variable: there, the share of each cohort that started a degree and left without one rose by 2.1 percentage points, from a baseline of 5 percent. The fact that dropout rose was not obvious a priori, since the reform could also have lowered it: a degree after three years let some students who would otherwise have left with nothing leave with a credential. Completion and dropout are both shares of the whole cohort, so for both to rise more people must have started. Taken together, this suggests that the reform increased enrolment, and that many of the students it drew in were unsure whether university would suit them: some found that it did and graduated, while others left without a degree. Section 6 formalises this mechanism.

Shorter first degrees could have meant less schooling: students who would have completed a four- or

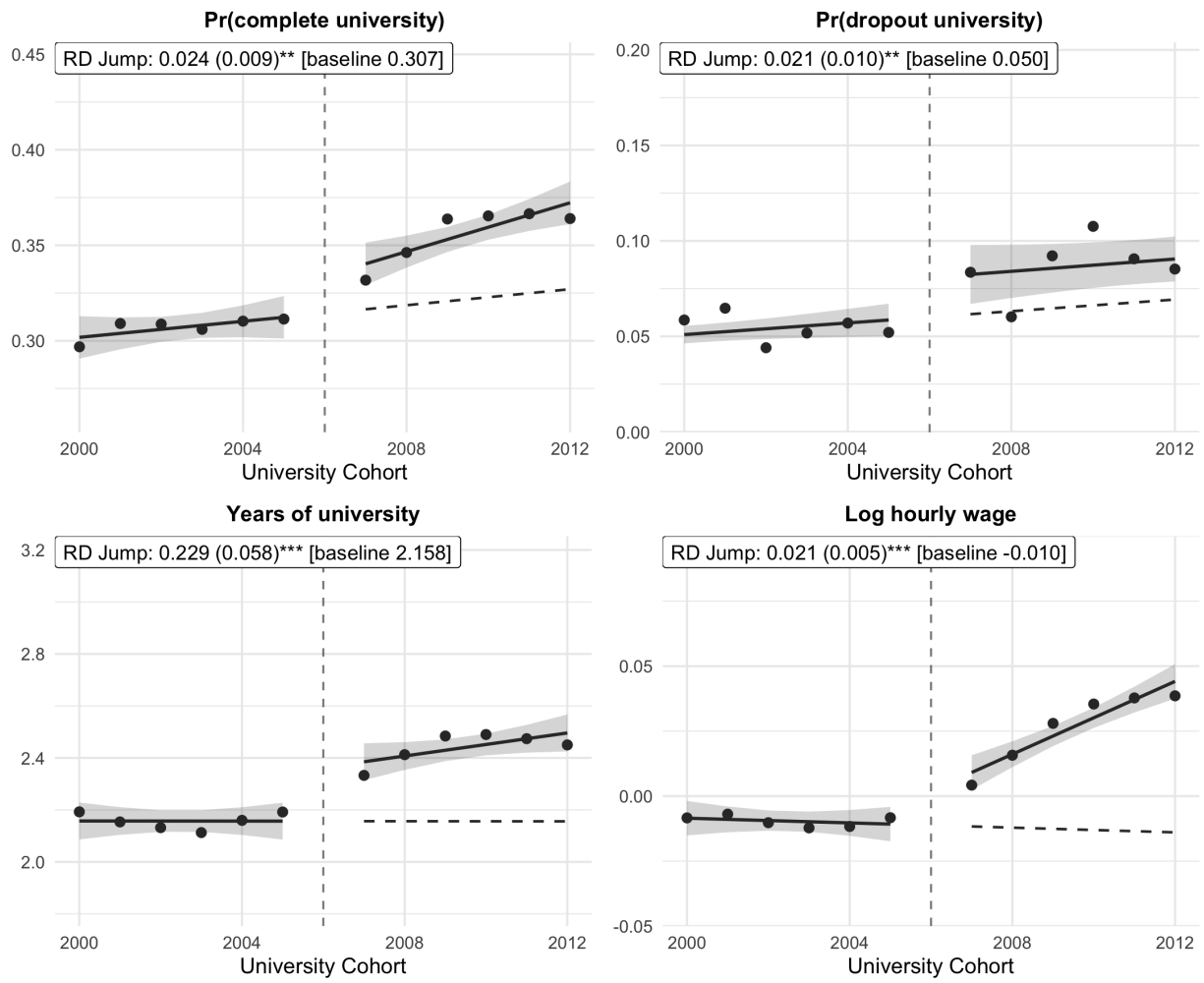


Figure 3: University completion, dropout, years of university and hourly wages, by cohort

Notes: Cohorts are defined by the year a person turned 18, 2000–2012; the 2006 cohort is excluded. Dots are cohort means; solid lines are the two lines of equation (2) with 95 percent confidence bands; dashed lines extend the pre-reform line. Boxes report the estimate of τ, its standard error and the baseline, the mean of the outcome in the pre-reform cohorts (Section 4). All graphs besides Pr(dropout university), which comes from IEFA data, come from *Quadros de Pessoal*. The bottom right graph shows the worker's mean of the log hourly wage relative to workers aged 40–45 in the same year (equation (12)), as in Table 1, column 1.

five-year degree could now stop after three. On net, however, schooling rose. Average years of university, counting zero for those without a degree, increased by 0.23 years, or 11 percent. The Labour Force Survey, which measures years of university directly from the age at which people left education, confirms the rise, showing a 16 percent increase (Appendix Table 12). The extra graduates more than made up for the shorter first degree, and, as Section 5.3 shows, more students, not fewer, went on to complete the full degree.

These gains carry over to the labour market. At ages 28 to 30, by which time most graduates have finished studying and started work, post-reform cohorts earn 2.1 percent more per hour (Table 1, column

1). Combined with the findings on years studied, this implies a return per year of university education of roughly 9%, slightly above the 7% usually estimated per year of compulsory schooling.

Wages are measured relative to workers aged 40 to 45 in the same year, to net out calendar year effects. The estimates are robust to different age windows, different cohort bandwidths, and placebo cutoffs (Appendices D.5 and G). I find no evidence that this came with a change in employment (Table 1, column 5). Overall, this provides evidence against what the critics feared.

Table 1: Effect of the reform on hourly wages at ages 28–30, the firm pay premium and employment

	Log hourly wage QP (1)	Firm effect QP (2)	Rest QP (3)	Firm share (%) (4)	Employed LFS (5)
All	0.021***	0.010***	0.011**	46	0.009
	(0.005)	(0.001)	(0.004)		(0.021)
Baseline	−0.010	—	—		0.759
Women	0.025***	0.013***	0.012*	53	0.027
	(0.006)	(0.002)	(0.005)		(0.026)
Baseline	−0.039	—	—		0.748
Men	0.015**	0.005***	0.009**	36	−0.011
	(0.004)	(0.001)	(0.004)		(0.018)
Baseline	0.020	—	—		0.771
p-value, women = men	0.18	0.00	0.70		0.22
Cohorts, pre / post	6 / 6	6 / 6	6 / 6		6 / 6
Observations	1,017,067	1,017,067	1,017,067		42,278

Notes: Estimates of τ in equation (2) at the 2007 cohort (Section 4). Columns 1–4: *Quadros de Pessoal*, one observation per worker: workers observed with an hourly wage at ages 28–30, main job, in the largest connected set of firms and workers; cohorts 2000–2012, 2006 excluded; the 2012 cohort is observed at ages 28–29 only. Column 1: the worker's mean, over the years in which they are aged 28–30, of the log real hourly wage minus the mean log hourly wage of workers aged 40–45 in the same year (equation (12)); baseline: pre-reform mean. Column 2: the worker's mean, over the same years, of the firm effect $\hat{\psi}_{J(i,t)}$ of equation (14) minus the mean firm effect of the employers of workers aged 40–45 in the same year (equation (15)). Column 3: column 1 minus column 2, so column 1 = 2 + 3 exactly. Firm effects are normalised, so columns 2 and 3 have no meaningful baseline. Column 4: column 2 as a percentage of column 1. Column 5 is a survey and keeps its own design: Labour Force Survey, everyone aged 28–30; employed (ILO definition); baseline: share employed in the 2005 cohort at ages 28–30. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To see how much of the gain comes from where people work, I decompose the wage into a worker effect, a firm effect and year effects (Abowd et al., 1999), and estimate the RD on the average firm effect of each worker's employers at ages 28 to 30 (Appendix D.6). I measure this firm effect relative to the employers of workers aged 40 to 45 in the same year, as I do for the wage. Because the firm effect and the rest of the wage are measured for the same workers in the same years, the RD on the wage equals the RD on the firm effect plus the RD on the rest. Column 2 of Table 1 shows that post-reform workers are at firms that pay 1.0 percent more to all their workers, relative to the firms of older workers. Sorting into better-paying firms thus accounts for 46 percent (column 4). The remaining 1.1 percent is also significant (column 3). Appendix F splits the gain by the type of firm people work in.

Cumulative earnings. The wage at 28 to 30 does not show what post-reform cohorts gave up while they studied longer. Measuring wages at younger ages would not show it either. At 23 or 25, some post-reform students are still in university, so a wage effect at those ages would mix the effect on pay with the later start of work, and the two could not be told apart. Cumulative earnings address this. Unlike the other outcomes, they are not one observation per worker at fixed ages, because they add up earnings over a range of ages. I sum each person's annual earnings from age 16 up to a given age, counting zero in years they are not observed in QP, and estimate the RD at each age. The sample is everyone who appears in QP at some point, so these totals are still conditional on eventually working in the private sector. The figure allows me to study two things: how much income post-reform cohorts forgo while they study, and whether the higher wages they earn once they start working make up for it. Figure 4 plots the estimates. In their early twenties, when more of them are still studying, post-reform cohorts fall behind. The gap is largest at age 25, about 4,900 euros. From then on post-reform cohorts earn more each year, and the gap closes around age 30. The higher wages thus offset the earnings forgone while studying by the early thirties.



Figure 4: Effect of the reform on cumulative earnings, by age

Notes: Quadros de Pessoal. An exception to the one-observation-per-worker design of Section 4: each point is a separate RD estimate of cumulative annual earnings from age 16 up to the age shown, in 2010 euros (consumer price index), with years not observed in QP counted as zero, and a 95 percent confidence band. Sample: everyone observed in QP at least once. Annual earnings are October monthly earnings times twelve, summed over jobs. RD on cohort means, linear on each side, cohorts 2000–2012 (2006 excluded). A cohort enters at an age only if it is observed at that age by 2023, so at 31 the post-reform side has four cohorts and at 32 three (2007–2009). Appendix C.1 describes the construction. Women and men separately reported in Appendix Figure 18.

Women and men. The reform raised attainment more for women than for men (Appendix Figure 17). Years of university rise by 0.36 years for women, from a baseline of 2.65, and by 0.13 years for men, from 1.68. Among the workers observed at both 28 and 29, the share with a degree rises by 3.4 percentage points for women, from a baseline of 39 percent, and by 1.7 percentage points for men, from 23 percent, significant only at the 10 percent level (Appendix Table 12). In levels, the effects on women are two to three times those on men, and "larger for women" refers to levels throughout this section. Relative to each sex's baseline the difference is smaller: years of university rise by 14 percent for women and 8 percent for men, and completion by 9 and 7 percent. The difference between women and men is significant for years of university ($p = 0.01$) but not for completion ($p = 0.20$). In IEFA, dropout rises by similar amounts for women and men, 1.8 and 2.2 percentage points, but due to small sample sizes neither estimate is significant on its own. Wages rise by 2.5 percent for women and 1.5 percent for men (Table 1). The larger effect for women matches their larger gains in years of university, but the difference between the two wage effects is not statistically significant ($p = 0.18$). The firm effect rises by 1.3 percent for women and 0.5 percent for men, and this difference is significant; it accounts for 53 percent of the gain for women and 36 percent for men. Cumulative earnings follow the same path for women and men (Appendix Figure 18).

5.2 Fields of study, sectors and occupations

The reform did not affect all fields in the same way (see Section2), and students responded to the differences. I group fields by what the reform did to the time to a first credential (Figure 1): fields where a five-year degree gained a three-year exit ("5 to 3+2", mostly engineering, architecture and computing); fields where a four-year first degree became three years ("4+2 to 3+2", economics and management, the sciences, the humanities and the arts); and fields with no usable early exit ("no change", health, including nursing). The outcome is the share of the whole cohort holding a degree in each group at age 28, so it combines students drawn into university with students who would have enrolled anyway but chose a different field. Figure 5 shows three results. First, degrees in fields that gained a three-year exit rise by 2.3 percentage points, a 26% increase from the baseline of 8.6 percent. Second, degrees in fields where the first degree became shorter rise by 1.6 percentage points, a 15% increase from the baseline of 10.7 percent. Third, degrees in the unchanged

fields fall, by 2.1 percentage points, about 30% from the baseline of 6.7 percent.

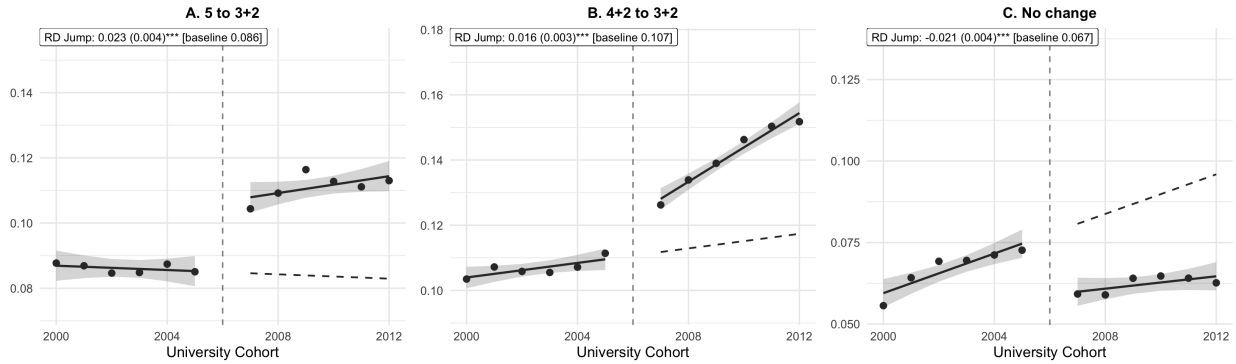


Figure 5: University degrees by what the reform did to the field, by cohort

Notes: Quadros de Pessoal, workers as for completion in Figure 3, one observation per worker. Share of the cohort whose highest completed level is a university degree in each group of fields.

Appendix Figure 26 splits the groups into fields. Most of the rise is in engineering and computing, whose degrees rise by 1.9 percentage points, from a baseline of 7.1 percent. Degrees in the sciences and mathematics rise by 0.4 percentage points. Economics and business degrees rise by 0.5 percentage points, from 5.3 percent. The humanities and the arts, and education, each rise by about 0.3 percentage points. The fall is in health, whose degrees drop by 2.1 percentage points, from 6.2 percent. Nursing stayed a four-year degree in Portugal and so became long relative to the alternatives. The rise in the two reformed groups, 4.0 percentage points, is larger than the overall rise in graduates, 2.4 percentage points, so the reform did more than add students: it also moved students who would have graduated anyway out of the unchanged fields. Other data sources show the same pattern (Appendix Table 13). Ministry records of applications and first-year students show the same shift at entry (Appendix J).

Sectors and occupations. The shift across fields carries into the jobs people hold. Figure 6 compares, for one field of each type, the jump in the share of the cohort holding a degree in the field with the jump in the share working in the matching occupation and in the matching sector, all at age 28. None of these outcomes conditions on holding a degree. In engineering, all three move together. In the fields whose first degree became shorter (business, education, the humanities and the arts), degrees rise by 0.9 points and work in the matching occupations by 1.3 points, but work in the matching sectors does not change significantly. In health, the share with a degree falls by 2.1 points, work in health occupations by 2.2 points and work in

health firms by 1.7 points. Appendix K measures the match directly at graduates' first jobs. In engineering, the share of new graduates whose first job matches their field does not change significantly. In economics and business, the occupation match does not fall, and the sector match is a few points lower but mostly not significantly so.

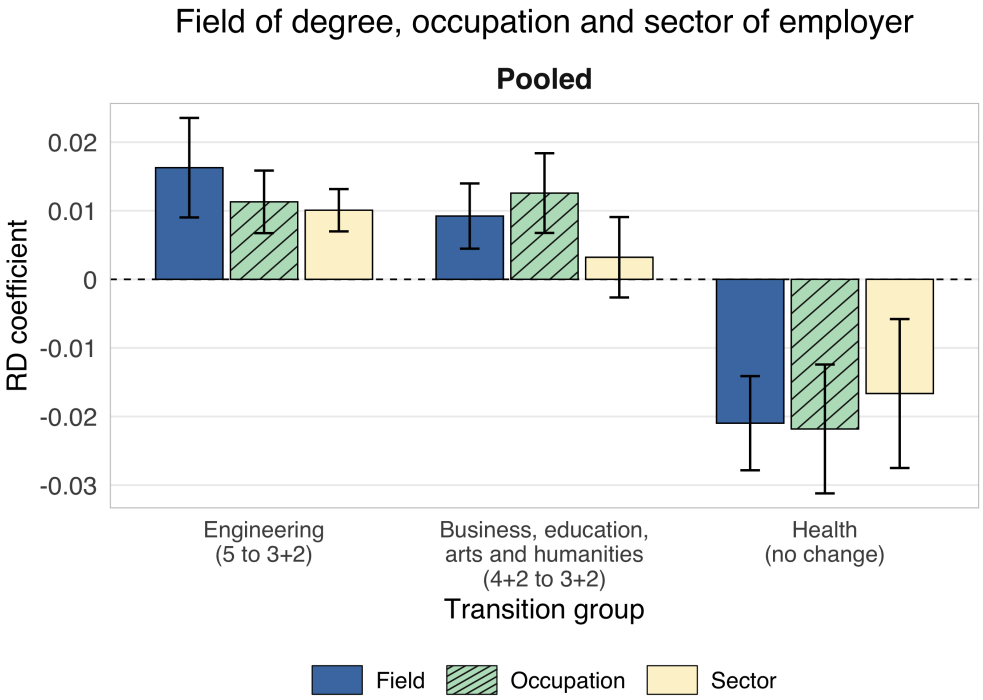


Figure 6: Field of degree, occupation and sector of employer, by what the reform did to the field

Notes: *Quadros de Pessoal*, workers observed at both 28 and 29, as in Figure 5, one observation per worker, cohorts 2000–2012 (2006 excluded). Estimates of τ in equation (2) with 95 percent confidence intervals. Each outcome is a share of the whole cohort on the worker's age-28 record; none conditions on holding a degree or on the sector. Field: the field of the highest degree as reported. Engineering: degree in engineering, manufacturing and processing, or architecture and construction (ISCED 52, 54, 58); occupation ISCO-08 214, 215, 251, 252 and 311; sector CAE 62, 63 and 71. Business, education, arts and humanities: degree in business and administration, education, or the humanities and the arts; occupation ISCO-08 231–235, 241–243, 262, 264 and 265; sector CAE 64–66, 69, 70, 85 (private schools only), 90 and 91. Health: degree in health (ISCED 72, with agriculture and veterinary science, ISCED 62 and 64); occupation ISCO-08 221–226 and 321–325; sector CAE 86.

Women and men. The field shifts are larger for women (Appendix Figure 19). Degrees in fields that gained a three-year exit rise by 2.9 percentage points for women, nearly half of their baseline of 5.9 percent, and by 1.7 percentage points for men, a seventh of their baseline of 11.2 percent. Degrees in fields where the first degree became shorter rise by 3.0 percentage points for women, a fifth of their baseline, and by 0.5 percentage points for men, which is not significantly different from zero. Degrees in the unchanged fields fall by 3.2 percentage points for women and 0.8 percentage points for men, in both cases nearly a third of

the baseline. The differences between women and men are significant at the 1 percent level in the last two groups and at the 10 percent level in the first. The LFS shows the same pattern. Degrees in the fields whose first degree became shorter rise by 3.7 percentage points for women and 0.9 percentage points for men, and health degrees fall, by 1.0 and 0.3 percentage points. Degrees in the fields that gained a three-year exit rise by 0.8 percentage points for women; for men the effect is small and not different from zero. Again, changes in occupations and sectors largely follow those of fields of study (Appendix Figure 20).

5.3 The intermediate qualification and the full degree

Engineering. Critics worried that students would spend less time in university, and so receive less training and earn lower wages. In engineering, this would happen if students took the new exit after three years instead of completing five. Engineering provides a direct test, in two data sources. The LFS asks graduates whether their degree is from before the reform, so it identifies five-year engineering degrees directly: a post-reform master's degree, or a pre-reform *licenciatura* (as explained in Section 3). In *Quadros de Pessoal*, occupations give an indirect measure. In Portugal the title of engineer, and the acts reserved for it, require the full five-year cycle, so an engineering occupation reserved for full-cycle graduates can only be reached by students who continued past the third year (Appendix B.2). If the reform had mainly diverted engineering students to the three-year exit, neither measure would rise. Both rise. In the LFS, the share of each cohort holding a five-year engineering degree rises by 1.1 percentage points, from a baseline of 2.9 percent (Figure 7, left panel). In QP, work in the engineering occupations that require the five-year degree rises by 0.7 percentage points, from a baseline of 3.1 percent (right panel). Engineering degrees at 28 rise by 1.4 percentage points. The reform brought students into engineering, and a fair share of them went on to the end.

Women and men. In the LFS, five-year engineering degrees rise by about 1 percentage point for both women and men (Appendix Figure 21). For women this is half their baseline of 2.0 percent; for men it is a quarter of their baseline of 4.0 percent, and significant only at the 10 percent level. In QP, work in the five-year engineering occupations at 28 to 30 rises by 0.9 percentage points for women, from a baseline of 1.9 percent. For men it rises by 0.4 percentage points, from 4.2 percent, which is not significantly different from zero. The difference between women and men in QP is not significant.

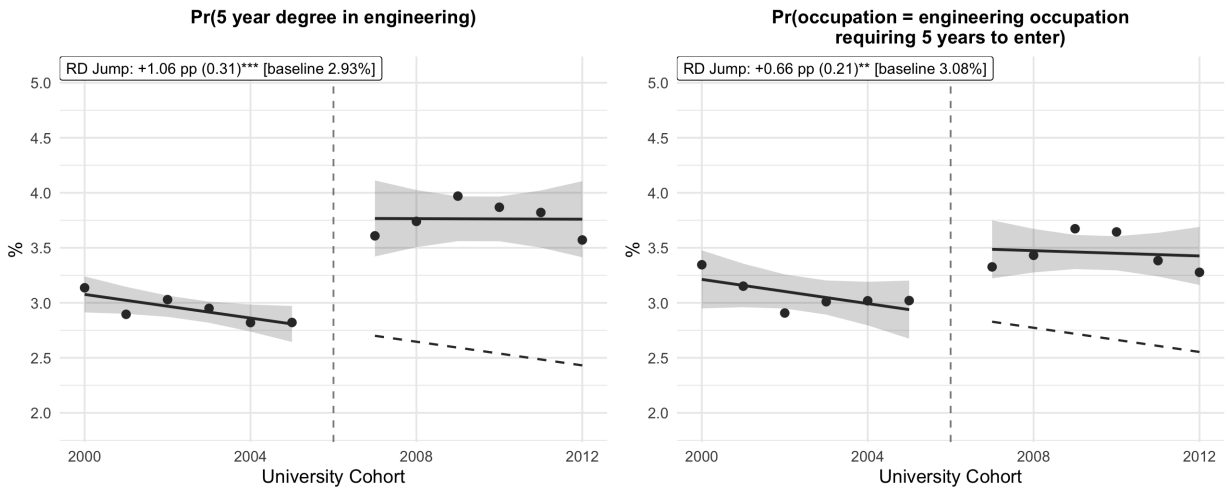


Figure 7: Five-year engineering degrees and five-year engineering occupations, by cohort

Notes: Left: Portuguese Labour Force Survey, a survey that keeps its own design. Outcome: the highest degree is in engineering, manufacturing and construction and takes five years: a master's degree or doctorate, or a *licenciatura* the respondent reports as from before the reform; percent of the whole cohort. Right: *Quadros de Pessoal*, percent of the cohort's in an engineering occupation that requires the five-year degree (ISCO-08 214, 215 and 216). See the notes to Figure 3.

5.4 Italy

Italy reformed its university system along the same lines six years earlier, and its students enter at a different age, so a Portuguese jump caused by something else that happened to the 2007 cohort would not reappear there. I therefore repeat each of the first three sets of results on Italian data.

Attainment and pay. Table 2 shows the main results. Average years of university rise by 0.13 years, from a baseline of 1.6 years, or 8 percent (column 2). The share holding a university degree rises by 1.4 percentage points, a 6% increase from the baseline of 23 percent (column 1). The log net monthly wage rises by 0.8 percent, significant only at the 10 percent level (column 3), and the probability of being employed does not change (column 4). Italian women gain more, as in Portugal. Average years of university rise by 0.17 years for women, or 9 percent, and by 0.07 years for men. The share of Italian women holding a university degree rises by 1.8 percentage points at the cutoff, and the share of men by 0.8 percentage points, compared with 2.4 and 0.9 percentage points in Portugal. Among Italian women, the log net monthly wage rises by 2.1 percent. Among men it does not change, and the difference between women and men is statistically significant. The Italian wage effect for women is close to the Portuguese one, 2.5 percent, although it is measured with a different wage concept, in a different country and six years earlier.

Table 2: Effect of the reform on university attainment, wages and employment in Italy

	Pr(university) (1)	Years of university (2)	Log net monthly wage (3)	Employed (4)
All	0.014***	0.125***	0.008*	−0.002
	(0.003)	(0.026)	(0.005)	(0.004)
Baseline	0.227	1.626	7.029	0.698
Women	0.018***	0.169***	0.021***	0.002
	(0.005)	(0.039)	(0.007)	(0.005)
Baseline	0.273	1.900	6.910	0.608
Men	0.008*	0.069*	0.001	−0.003
	(0.004)	(0.035)	(0.005)	(0.005)
Baseline	0.179	1.340	7.134	0.791
p-value, women = men	0.17	0.06	0.02	0.55
Observations	269,026	269,026	142,599	269,026

Notes: RD estimates at the year of adoption. Italian Labour Force Survey. The running variable is the year a person turned 19 minus the adoption year in their region; cohorts −4 to 3. Individual-level regression with a line on each side of the cutoff, the coefficient being the jump at the cutoff. Column 1: holds a university degree. Column 2: years of university, zero without a degree and age at the end of study minus 19 with one. Column 3: log net monthly pay in the main job, top-coded at 3,000 euros, employees only. Column 4: employed. Baseline: mean of the pre-reform cohorts. Observations (women / men): columns 1, 2 and 4, 137,193 / 131,833; column 3, 66,675 / 75,924. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Fields. Figure 8 puts the two countries side by side, with Italian fields grouped by the same rule. The Italian field estimates are for women and men separately. In Italy, degrees in the fields that gained a three-year exit rise by 1.2 percentage points for women and 0.7 percentage points for men; degrees in fields whose four-year degree became three plus two rise by 0.9 percentage points for women and do not change for men. The Italian field effects are roughly half the Portuguese ones, and in both countries they are larger for women. The second group, however, is not a direct counterpart of its Portuguese equivalent. In Italy the four-year *laurea* had no master's above it, so the reform did two things at once: it brought the first degree forward by a year, and it made that degree the first step towards a five-year qualification. The old four-year degree was also equated in law with the new master's, not with the three-year degree (Appendix B.2). A student who wanted a qualification of the old standing therefore faced a longer path after the reform. This may explain why the response in this group is smaller in Italy than in Portugal, and absent among men. Healthcare offers a contrast with Portugal. In Italy nursing was already a three-year diploma, and the reform turned it into a first degree with a master's on top. Nursing therefore did not become long relative to other fields, as it did in Portugal; instead it gained a route to further study. Healthcare degrees rise by 0.8 percentage points among women and 0.3 percentage points among men (Table 3). In both countries, then, students moved towards the fields where the reform added a credential and away from those where it did not.

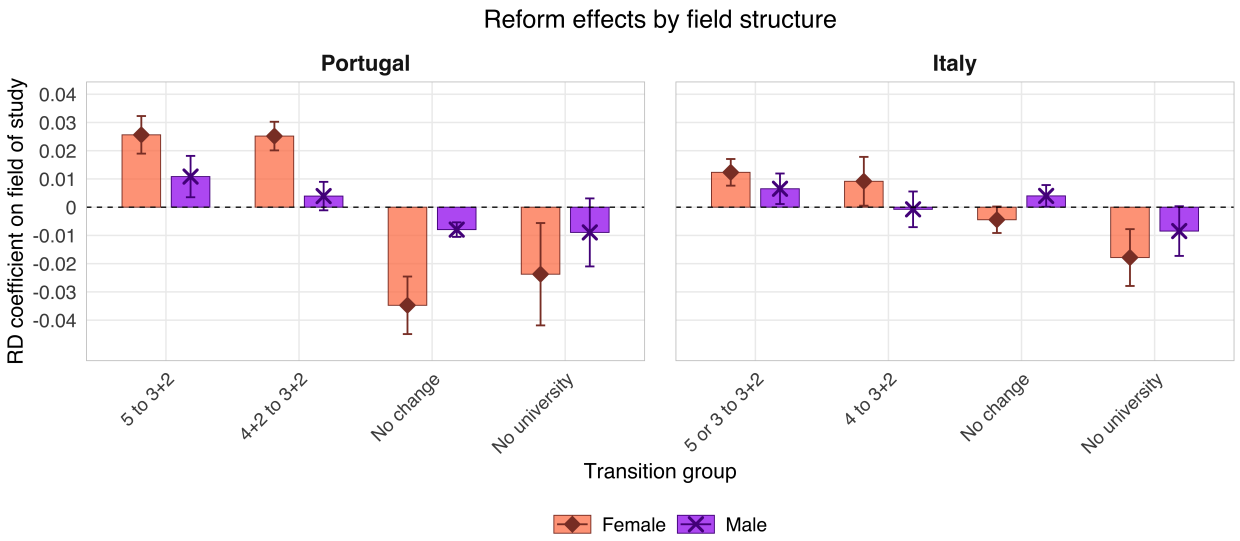


Figure 8: Effects on degrees by what the reform did to the field, Portugal and Italy

Notes: RD estimates with 95 percent confidence intervals, women and men.

Long degrees. The Italian survey separates degrees of four or more years from shorter ones, so it can check whether students stopped at three years. Table 3 shows this did not happen. Engineering degrees rise by 0.4 percentage points, from a baseline of 2.7 percent, and engineering degrees of four or more years by 0.2 percentage points, significant only at the 10 percent level. Among women, engineering degrees rise by 0.4 percentage points, and engineering degrees of four or more years by 0.3 percentage points, so most of the new engineering graduates completed a long degree. Among men the engineering estimates are similar in size, 0.4 and 0.2 percentage points, but they are not statistically different from zero. The share completing a 5-year degrees in healthcare rises by 0.4 percentage points among women and 0.2 percentage points among men.

Overall, the reform raised the number of graduates, years of university and pay. How it did so rules out two simple explanations. If the reform had only made degrees cheaper, the five-year engineering degree, which did not become any shorter, could only have lost students to the cheaper three-year exit. Instead, more students completed it. If the new exit had mainly given a degree to students who would otherwise have left with nothing, dropout would have fallen. Instead, it rose. Both facts fit one explanation: students do not know, when they enrol, whether university will suit them, and they find out by trying. The three-year exit lowered what a student loses if she tries and fails: failing after the third year now leaves her with a

Table 3: Engineering and healthcare degrees by length, Italy

	Engineering		Healthcare	
	Any degree	5-year degree	Any degree	5-year degree
All	0.004***	0.002*	0.006***	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)
Baseline	0.027	0.022	0.022	0.008
Women	0.004***	0.003**	0.008***	0.004***
	(0.001)	(0.001)	(0.002)	(0.001)
Baseline	0.012	0.009	0.030	0.009
Men	0.004	0.002	0.003*	0.002**
	(0.002)	(0.002)	(0.001)	(0.001)
Baseline	0.043	0.034	0.015	0.006
p-value, women = men	0.918	0.725	0.022	0.113
Observations	269,026	269,026	269,026	269,026

Notes: Italian Labour Force Survey. The running variable is the year a person turned 19 minus the adoption year in their region. RD estimates from individual-level regressions with separate linear trends on each side of the cutoff. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Baseline is the mean over pre-reform cohorts. "Any degree": any university degree in the field. "5-year degree": a degree of 5 years (single-cycle or *laurea specialistica*) in the field. "All" pools women and men in one regression. Observations: 269,026 in all, 137,193 women and 131,833 men.

degree instead of nothing. This is worth most to students who are unsure, so the exit drew them in. Most of the new students whom university did not suit found out early, before reaching the exit, and left with nothing, so dropout rose. Those who passed their first years grew more confident, and many went on to the end, so more students completed the five-year degree. The shorter first degree worked through cost instead: it saved a year for every student who finished, so it mattered most to students who already expected to finish. Both changes made the reformed fields more attractive, so students moved away from the unchanged ones. Section 6 turns this explanation into a model and estimates it.

6 A model of university choice

The goal of this section is to explain why the option to leave with an intermediate credential after three years increases completion of the full five-year degree. In theory, this option can generate both positive and negative effects on five-year completion. On the one hand, students who would have previously completed the full five-year degree may now stop at the intermediate credential. On the other hand, the intermediate credential will attract new students, some of whom may complete the full five-year degree. The reduced-form alone can tell me which effect dominates, but it cannot separately identify the two forces.

I therefore develop a model of education as experimentation (Manski, 1989) that allows me to quantify each force and simulate counterfactual reforms. The model matches my reduced-form results with a suitably

selected parameter vector. It predicts that (i) the reform increases university completion and dropout; (ii) post-reform cohorts are more likely to complete fields that shortened their first degree and fields that gained an intermediate credential; and (iii) the option to leave with an intermediate credential after three years increases full five-year completion.

I disentangle the positive and negative effects of intermediate credentials by treating enrolment in university as a risky choice. Students decide to enrol based on uncertain beliefs about how likely they are to graduate. Adding an intermediate credential lowers this risk of attending university by making it easier to graduate. Thereby, it attracts additional students with lower beliefs. Some of those students do better than expected and complete even the full degree. On the other hand, students whose beliefs were barely high enough to enrol in the five-year degree beforehand may switch to a three year degree. This can happen if the pay difference between three and five year degrees is low enough. Students with a sufficiently low belief may find it not worthwhile to complete the full five-year degree.

6.1 The Model

When agents graduate high-school, at $t = 0$, they face the decision of whether to enter the labour force or enrol in university. Agents differ in their match with university, θ, which is either good (G) or bad (B). Good matches never fail, while bad fail every year with some probability which I describe later. The match θ is not observable by the agent. Instead, a high school graduate, when deciding whether to enrol in university, has a belief about her type, denoted by $p_0 \in [0, 1]$, where $p_0 = \Pr(\theta = G)$, and this belief is correct on average.

At any period in time, an agent can either be working (W), studying in a field that gained an intermediate credential with the reform (IC), studying in a field whose first degree was shortened (S), or studying in a field where nothing changed (N). Let $k \in K = \{IC, S, N\}$ denote the field of study. One period is a year, and the direct cost of studying is c per year of study in every field. Let t denote the number of years the student has passed. She graduates from field k when she has passed $\bar{t}_k$ years. In IC, a student can graduate after five years both before and after, but after the reform she can also graduate after three years. In S, a student can graduate after 4 or 6 years before the reform, and after 3 and 5 years after it. In N, $\bar{t}_N = 4$ before and after.

Work is an absorbing state: an agent who starts working never returns to university, and her yearly pay stays constant until her working life ends. An agent who never enrols earns w_0, which I normalise to one. A student who leaves field k after passing t years earns d_t^k:

$$\text{Before the reform:} \quad d_t^{IC} = \begin{cases} w_0 & \text{if } t < 5 \\ w_5^{IC} & \text{if } t = 5 \end{cases} \qquad d_t^S = \begin{cases} w_0 & \text{if } t < 4 \\ w_4^S & \text{if } t = 4, 5 \\ w_F^S & \text{if } t = 6 \end{cases} \qquad d_t^N = \begin{cases} w_0 & \text{if } t < 4 \\ w^N & \text{if } t = 4 \end{cases}$$

$$\text{After the reform:} \quad d_t^{IC} = \begin{cases} w_0 & \text{if } t < 3 \\ w_3^{IC} & \text{if } t = 3, 4 \\ w_5^{IC} & \text{if } t = 5 \end{cases} \qquad d_t^S = \begin{cases} w_0 & \text{if } t < 3 \\ w_3^S & \text{if } t = 3, 4 \\ w_F^S & \text{if } t = 5 \end{cases} \qquad d_t^N \text{ as before,} \tag{4}$$

with $w_0 < w_3^{IC} < w_5^{IC}$, $w_0 < w_4^S < w_F^S$, $w_0 < w_3^S < w_F^S$ and $w_0 < w^N$. That is, in every field a credential pays more than no credential, and the last credential pays more than an earlier one. The three-year degree in S may pay less than the old four-year degree. The reform affects pay in two ways. In field IC students can reach w_3^{IC} if they pass 3 years rather than falling back to the lowest pay w_0. In field S students reach the intermediate pay one year earlier, but earn less (w_4^S versus w_3^S).

Assumption 1. *Pay depends on the credential and not on the match: d_t^k is the same for $\theta = G$ and $\theta = B$.*

Employers don't observe the type θ and pay only based on the completed education credential, consistent with Stange (2012) and Arcidiacono et al. (2025).

While she studies, a student receives a signal at the end of each year: she either passes or fails. A student who passes moves from t to $t + 1$, a student who fails drops out and enters the labour market.

Assumption 2. *In every field and year,* $\Pr(\textit{fail} \mid \theta = G) = 0$ *and* $\Pr(\textit{fail} \mid \theta = B) = q$ *with* $q \in [0, 1]$.

A good match never fails, so passing automatically increases beliefs. Trachter (2015) imposes the same restriction on high-ability students at two-year colleges.

Assumption 3. *Beliefs are correct on average: among agents with belief p_0, a share p_0 are good matches.*

Upon completing high-school, an agent chooses between work and enrolling in a field. If she enrols, she then chooses every year whether to continue. She makes these choices to maximise the expected utility of

her lifetime pay. A student who studies for s years and then works at yearly pay d has lifetime pay

$$Y(s,d) = \sum_{a=s}^{T-1} \beta^a d \; - \; \sum_{a=0}^{s-1} \beta^a c, \tag{5}$$

where a counts years since age 18, T is the length of working life and β is the discount factor. A year of study therefore costs c plus the pay she gives up. The utility function is $u(Y) = Y^{1-\gamma}/(1-\gamma)$, where γ is the coefficient of relative risk aversion. Preferences are defined over the present value of lifetime pay, so the model has no consumption or saving decision and no credit constraint (Trachter, 2015).

Let $V_t^k(p_t)$ denote the value for a student in field k who has passed t years and holds belief p_t, and let $u(Y(t, d_t^k))$ denote the value for a worker who leaves at that point. I measure the value of enrolling in units of lifetime pay, using its certainty equivalent $v_k(p_0) = u^{-1}\big(V_0^k(p_0)\big)$: the sure lifetime pay the agent values as much as the gamble of enrolling in k. Work carries no risk, so $v_W = Y(0, w_0)$. At 18 each agent also draws a preference shock ε_k for each option, independent across options and distributed Type I extreme value with location zero and scale σ. She solves

$$\max_{k \in \{W,IC,S,N\}} \big\{ v_k(p_0) + \mu_k + \varepsilon_k \big\}, \qquad \Pr(k \mid p_0) = \frac{\exp\big[(v_k(p_0) + \mu_k)/\sigma\big]}{\sum_{j \in \{W,IC,S,N\}} \exp\big[(v_j(p_0) + \mu_j)/\sigma\big]}, \tag{6}$$

with $\mu_W = 0$. The parameter μ_k is the mean preference for field k relative to work, in years of pay without a degree. The scale parameter σ governs how sensitive the choice of a field is to its value.

Assumption 4. *Preference shocks are drawn once, at 18, independently of the match, and they enter only the choice of field.*

The problem of a student

The student updates her beliefs after receiving the signal. After passing year $t+1$, she updates it by Bayes' rule:

$$p_{t+1} = \Pr(\theta = G \mid \text{pass}, p_t) = \frac{\Pr(\text{pass} \mid \theta = G)\, \Pr(\theta = G \mid p_t)}{\Pr(\text{pass} \mid p_t)} = \frac{1 \cdot p_t}{p_t + (1-p_t)(1-q)}. \tag{7}$$

After t passed years her belief is $p_t = p_0/[p_0 + (1-p_0)(1-q)^t]$. Her belief also determines the chance she thinks she will fail, because she does not know her match: $\Pr(\text{fail} \mid p_t) = \Pr(\text{fail} \mid \theta = G)\Pr(\theta = G \mid p_t) + \Pr(\text{fail} \mid \theta = B)\Pr(\theta = B \mid p_t) = (1-p_t)\,q$. If $t < \bar{t}_k$, she chooses between leaving with d_t^k and

continuing for one more year, and her value is

$$V_t^k(p_t) = \max\Big\{ \underbrace{u\big(Y(t, d_t^k)\big)}_{\text{leave}},\ \underbrace{\overbrace{(1-p_t)\,q}^{\Pr(\text{fail}|p_t)}\ u\big(Y(t+1, d_t^k)\big) + \overbrace{\big[1-(1-p_t)\,q\big]}^{\Pr(\text{pass}|p_t)}\ V_{t+1}^k(p_{t+1})}_{\text{continue}} \Big\}. \tag{8}$$

If $t = \bar{t}_k$, she graduates and receives $u(Y(\bar{t}_k, d_{\bar{t}_k}^k))$. A student who enrols commits to the first year, so the value of enrolling is $V_0^k(p_0) = (1-p_0)\,q\,u(Y(1, w_0)) + [1-(1-p_0)\,q]\,V_1^k(p_1)$. Figure 9 shows what happens every year.

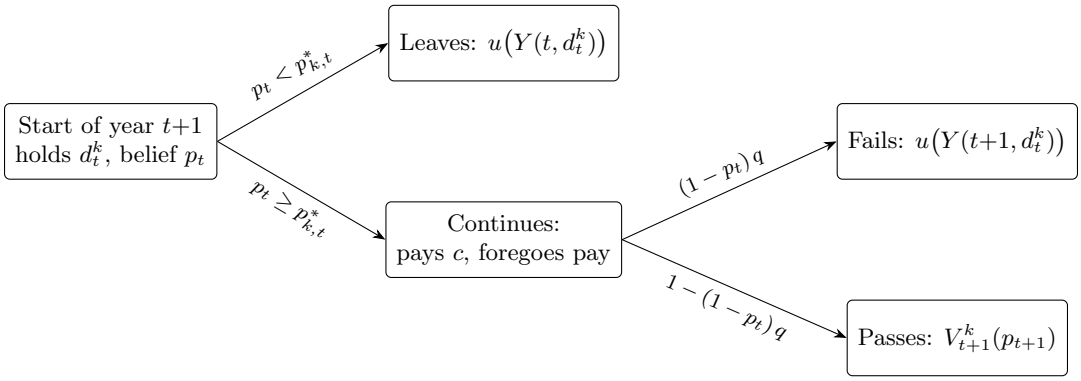


Figure 9: Timing of a year of study

A student stays at university only if she confident enough that she will pass, i.e. if $p_t \geq p_{k,t}^*$. Otherwise, she will leave. This is because the value of staying at university rises with p_t, as a higher p_t lowers the chance of failing. The alternative decision to leave university does not depend on p_t.

The reform changes the student problem in two ways, each affecting a different component of equation (8). The intermediate credential raises d_3^{IC} and d_4^{IC}. Every plan available before the reform remains available at weakly higher pay, so the value of enrolling in IC cannot fall at any belief. A student who fails in year four or five, or who stops after year three, now leaves with w_3^{IC} instead of w_0, so the credential is worth most to students with low beliefs. Risk aversion raises its value. If w_3^{IC} is high enough, even a student sure of a good match stops at three: this happens when $Y(3, w_3^{IC}) \geq Y(5, w_5^{IC})$.

The shorter first degree brings both credentials forward by one year, which saves a year of cost and forgone pay and lowers the chance of failing before the first credential. But the three-year degree may pay less than the old four-year degree, $w_3^S < w_4^S$. A low confidence student can lose, because there no longer is an exit after four years: she must either stop with w_3^S or risk two more years of study. Unlike the intermediate credential, the shorter degree can therefore lower the value of the field.

Dropout can either rise or fall. It may rise because the reform draws in more students with low beliefs, who fail at higher rates. It may fall because some students who would have left with no credential now leave with a credential.

6.2 Estimation

I estimate the model by minimum distance, using the reform effects of Section 5 as moments. The vector of estimated parameters is

$$\psi = \left(\mu_{IC},\ \mu_S,\ \mu_N,\ q,\ \sigma,\ w_3^{IC},\ w_3^S/w_4^S\right), \tag{9}$$

which contains the three mean preferences ($\mu_{IC},\ \mu_S,\ \mu_N$), the failure probability q, the scale parameter σ, the bachelor's pay in field IC (w_3^{IC}), and the pay of the three-year first degree relative to the old four-year one in field S (w_3^S/w_4^S). Table 4 lists each parameter and the moment that identifies it.

The estimation minimises the distance between the model and the data, subject to the model exactly matching the share of the cohort holding a degree in each field group before the reform:

$$\hat{\psi} = \arg\min_{\psi} \sum_{m=1}^{5} \left(\frac{\hat{x}_m(\psi) - x_m}{\mathrm{SE}_m} \right)^2 \quad \text{subject to} \quad \hat{x}_k^{\mathrm{pre}}(\psi) = x_k^{\mathrm{pre}},\ \ k \in \{IC, S, N\}, \tag{10}$$

where x_m is a data moment, $\hat{x}_m(\psi)$ is its counterpart in the model, SE_m is the standard error of the data moment, and x_k^{pre} is the share holding a degree in field k before the reform (Appendix L.1).

The sum runs over five moments, in percentage points of the cohort: dropout before the reform and four RD effects. These are the jumps in the share holding a degree in IC and in S (Figure 5), the jump in dropout (Figure 3), and the jump in work in the engineering occupations that require the five-year degree (Figure 7, right panel). The last is a lower bound on the rise in five-year degrees in IC, because not every five-year graduate works in these occupations, so it counts only when the model falls below it. Together with the three shares before the reform, this gives eight moments for seven parameters, so the model is overidentified by one moment.

The remaining parameters are set outside the estimation (Appendix Table 17). I set the discount factor to $\beta = 0.95$, as in Stange (2012), and the coefficient of relative risk aversion to $\gamma = 2$, as in Johnson (2013). Working life lasts $T = 40$ years. The direct cost of a year of study is $c = 0.1$, close to the legal maximum fee

in public universities as a share of the yearly pay of a worker without a degree. The pay of the old degrees comes from *Quadros de Pessoal*, relative to secondary-school leavers of the same age and year: $w_5^{IC} = 1.6$, $w_4^S = 1.4$, $w_F^S = 1.5$ and $w^N = 1.4$.

Starting beliefs are also set outside the estimation. Using PISA 2003 for Portugal, whose students reached university age just before the reform, I predict the chance that each student expects to finish university from her test scores and her confidence. I summarise these predictions with a Beta(0.94, 0.91), whose mean is 0.51 (Appendix L.1).

Table 4: Estimates, the targeted moments that identify them, and untargeted moments

Parameter	Estimate	Targeted moment	Data	Model	Untargeted moment	Data	Model
Mean preference: μ_{IC}	−1.74	Degree in IC, before	8.61	8.61	Δ degree in N	−2.08	−1.18
Mean preference: μ_S	−0.62	Degree in S, before	10.67	10.67	Δ five-year degree, LFS	+1.06	+0.66
Mean preference: μ_N	−0.77	Degree in N, before	6.67	6.67	Bachelor's pay, % of five-year	86.8	87.4
Failure probability: q	0.412	Dropout, before	4.97	5.00	Δ mean log pay	+2.08	+1.82
Scale parameter: σ	0.321	Δ degree in IC	+2.33	+2.37			
Bachelor's pay: w_3^{IC}	1.3988	Δ five-year occupation	≥ 0.66	+0.66			
Three-year pay: w_3^S/w_4^S	0.898	Δ degree in S	+1.63	+1.67			
Overidentifying moment		Δ dropout	+2.08	+0.54			

Notes: Each parameter sits beside the targeted moment that mainly identifies it; the untargeted moments on the same row are not linked to that parameter. Mean preferences and scale parameter in years of pay without a degree ($\mu_W = 0$). Moments in percent of the cohort, except bachelor's pay and mean log pay (log points). Standard errors: Appendix L.1 for the estimates, Appendix Figure 27 for the data. Distance at the estimates: 2.49.

Each parameter is identified mainly by one moment, although the estimation is joint, and Table 4 lists each parameter next to that moment. The mean preferences are identified by the field shares before the reform. The failure probability is identified by dropout before the reform: the more often bad matches fail, the more students leave with nothing. The scale parameter is identified by the jump in degrees in IC: given pay and beliefs, the model gives the rise in the value of enrolling in IC, and σ is the size of the preference shocks that makes entry respond as much as the data show. The bachelor's pay is identified by the five-year jump: the higher it is, the more students stop at three. The pay of the three-year first degree is identified by the jump in S. The jump in dropout provides the overidentifying moment.

Table 4 reports the estimates. A bad match fails with probability 0.41 each year, so two in three bad matches have failed by the end of the second year. The model matches seven of the eight targeted moments, including the rise in degrees in both reformed groups and the rise in five-year degrees. It also gets the effect on dropout right in sign, although too small in magnitude. The estimated bachelor's pay is 87.4 percent of five-year pay, close to the 86.8 percent in *Quadros de Pessoal*. The model also predicts a rise in the cohort's

mean log pay of 1.82 log points, within one standard error of the 2.08 (0.54) in the data (Table 1, column 1).

The two effects on five-year completion

As explained above, an intermediate credential can have two opposing effects on full five-year completion. I use the model to quantify these two effects separately, as shown in Table 5. The intermediate credential draws in students who finish five years and would not have done so before the reform: 1.3 percent of the cohort. It also leads some students to stop at three who would have finished five: 0.5 percent of the cohort. A further 0.1 points move to other fields, so five-year degrees rise by 0.66 points on net.

Table 5: Flows into and out of the five-year degree in IC in the estimated model

	Percent of the cohort
Five-year degree before the reform	8.61
Not in IC before, five-year degree after	+1.33
Five-year degree before, stop at three after	−0.54
Five-year degree before, another field after	−0.13
Five-year degree after the reform	9.27
Net change	+0.66

Notes: Flows in the estimated model. A student keeps her starting belief, match, preference shocks and the year she would fail before and after the reform.

6.3 Counterfactuals

I use the estimated model to ask three questions. Do students need to learn for the model to explain the responses? How do the answers depend on the pay gap between the bachelor's and the five-year degree? And when should the intermediate credential come?

Do students need to learn for the model to explain the responses? In the model a student learns about her match with university by passing or failing. A model without learning gives student the same risk of failing but no learning, so passing does not update her belief and she keeps her starting belief p_0 throughout. A student fails each year with the same chance $r(p_0)$, whether she is a good or a bad match, set so that her chance of passing five years is the same as with learning: $[1 - r(p_0)]^5 = p_0 + (1 - p_0)(1 - q)^5$. The two models differ only in when failure happens. With learning, the chance of failing falls as a student passes: for a student with the average starting belief, 0.51, it is 20 percent in the first year and 4 percent in

the fifth. Without learning, the chance of failing is 12 percent in every year. Failures therefore come later: a third of them after the third year, against a seventh with learning. This has important consequences for the field gaining an intermediate credential, shown in Table 6. At the estimates, the model without learning overstates the response: degrees in IC rise by 10.8 percentage points of the cohort, against 2.3 in the data. This is because the probability of failing in the first years is lower, which attracts more students, and because more failures come after the three-year credential, so the credential insures more people. At the same time, students reach year three as unsure as when they started, so five-year degrees fall by 1.6 points instead of rising. Re-estimated, the model without learning matches the rises in degrees in IC and S and the five-year bound, but only with a bachelor's that pays 72 percent of five-year pay, far below the 86.8 percent in *Quadros de Pessoal*. With the bachelor's pay held at the QP ratio, it fits much worse, with a distance of 17.4, against 2.67 for the model with learning re-estimated at the same ratio (Appendix L.2). Learning is what lets the model match the responses at the bachelor's pay seen in the data.

Table 6: The model with and without learning

			No learning		
	Data	Estimated model	At the estimates	Re-estimated	Re-estimated, QP pay ratio
Δ degree in IC	+2.33	+2.37	+10.82	+2.41	+2.95
Δ degree in S	+1.63	+1.67	−0.81	+1.69	+1.65
Dropout, before	4.97	5.00	4.27	4.93	3.89
Δ dropout	+2.08	+0.54	+0.92	−0.31	−0.32
Δ five-year degree in IC	$\geq$ 0.66	+0.66	−1.64	+1.76	+0.41
Δ degree in N (check)	−2.08	−1.18	−2.59	−0.49	−0.33
Bachelor's pay, % of five-year pay	86.8	87.4	87.4	72.3	86.8
Distance		2.49	707.0	6.01	17.4

Notes: Effects in percentage points of the cohort; data as in Table 4. No learning: each student fails each year with the same chance, set so that her chance of passing five years is the same as in the estimated model (Appendix L.2); average tastes set again to match the shares before the reform. Re-estimated: the same search as the estimated model; in the last column the bachelor's pay is held at its ratio to five-year pay in *Quadros de Pessoal* (data column). The estimated model re-estimated at that ratio has a distance of 2.67.

The pay gap between the intermediate credential and the full five-year degree. Critics of the reform feared that adding an intermediate credential would replace the five-year degree instead of insuring it. In the model, this depends on the pay gap between the two. Panel (a) of Figure 10 varies the bachelor's pay, as a share of five-year pay. The edge at which even a student sure of a good match stops at three is at 87.8 percent of five-year pay: the five-year degree must pay at least 14 percent more than the bachelor's

for a sure good match to finish it. Below the edge, the bachelor's insures. Up to about 86.5 percent, almost nobody who would have finished five years stops at three, and the share completing five years rises with the bachelor's pay, because a better-paid bachelor's draws in more unsure students, some of whom turn out to be good matches and finish. The rise peaks at 1.9 percentage points of the cohort, at 86.5 percent. Closer to the edge, students who are still unsure after three years start to stop. At the estimate, 0.5 percent of the cohort stop at three who would have finished five years before the reform, while 1.3 percent finish five years who would not have before. From 87.5 percent, five-year degrees fall below their level before the reform, and from 87.8 percent nobody finishes five years. The estimate, 87.4 percent, and the ratio in *Quadros de Pessoal*, 86.8 percent, both sit in this narrow band just below the edge. Students in the model stay for five years because the pay gap is just large enough.

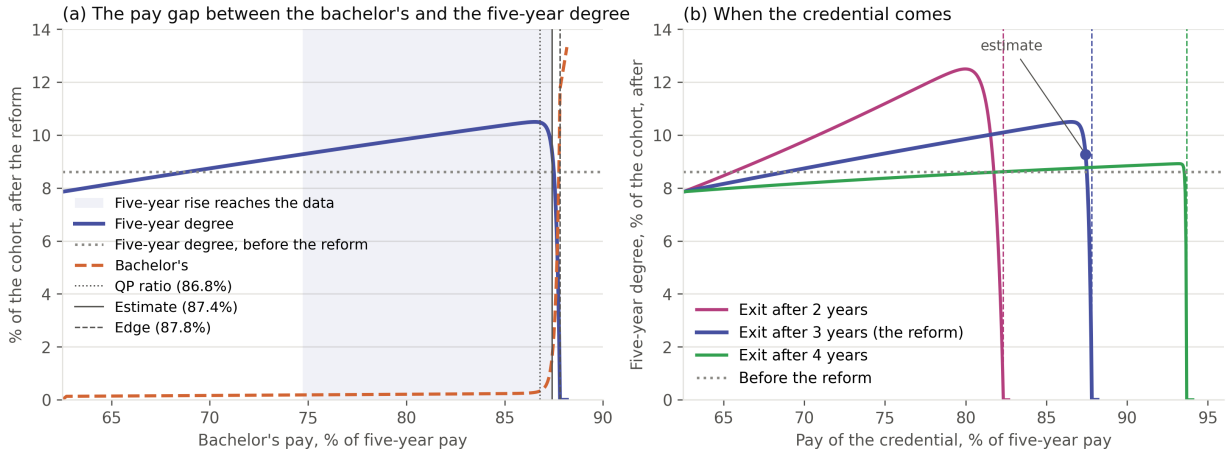


Figure 10: The pay and the timing of the intermediate credential in the estimated model

Notes: Share of the cohort with each credential after the reform, in percent, with both changes of the reform in place and everything else at the estimates. (a) As the bachelor's pay varies, as a share of five-year pay. Shaded: the range in which the five-year rise reaches the data's lower bound, 0.66 points. Vertical lines: the ratio in *Quadros de Pessoal*, the estimate, and the edge, above which a student sure of a good match stops at three. (b) Five-year degree as the pay of the credential varies, for a credential after two, three (the reform) and four years. Dashed vertical lines: the edge of each credential. Dot: the estimated model.

How does 3+2 compare to 2+3 and 4+1? Panel (b) of Figure 10 moves the intermediate credential to after two or four years (see Appendix L.3 for details). The timing trades insurance against competition with the full degree. An earlier credential insures more students: of the bad matches who enter, 28 percent fail after passing two years, 13 percent after passing three, and 5 percent after passing four. But an earlier credential competes with the full degree at a lower pay. Its edge is at 82 percent of five-year pay for a

credential after two years, 88 percent after three, and 94 percent after four. A credential after four years changes little at any pay below its edge: it insures few students, so it raises degrees in IC by at most 0.4 points. If it paid the same premium per year of study as the estimated bachelor's, it would pay more than its edge, and nobody would finish five years. The earlier the credential, the more it insures, and the less it can pay before it competes with the full degree.

7 Conclusion

When Europe launched the Bologna Process, its advocates expected shorter and more flexible degrees to widen access to university. Its critics expected students to spend less time at university and, as a result, to earn less. The evidence from Portugal and Italy favours the advocates. More students graduated. On net, post-reform cohorts spent more years at university, and by their late twenties they earned more.

The reform shortened degrees in some fields and introduced an intermediate exit qualification in others. The model shows that these changes operate through different channels. A shorter first degree lowers the cost of studying. It saves a year for every student who finishes and therefore benefits most those already likely to complete the degree. An intermediate qualification instead provides insurance. It changes little for a student who completes the full programme, but it makes enrolling less risky. A student who discovers that the field does not suit her can leave with a recognised qualification rather than nothing. This option is valuable to students who are uncertain at 18, whether or not they ultimately use it, and it is these students whom flexibility draws in. A country can therefore make a long programme less risky to enter without making it any shorter, and adding an intermediate qualification can increase completion of the full degree.

Two conditions made this work in Portugal. First, the full degree retained its value. The title of engineer still required all five years, giving students who turned out to be well matched a reason to stay, and many did. My counterfactuals show that for this to happen the intermediate qualification has to be valuable enough to make enrolment less risky, but not so valuable relative to the full degree that well-matched students chose to leave early. Second, employers absorbed the additional graduates. The composition of occupations and sectors shifted in parallel with the composition of fields of study, and much of the wage gain came from employment at higher-paying firms. The value employers place on shortened degrees matters too. If

employers discount them too heavily relative to the old degrees, the lower return can outweigh the benefit of finishing sooner, and enrolment may fall rather than rise. Where employers cannot absorb additional graduates in a field, the same reform may raise attainment without raising pay.

These lessons extend beyond Europe and are relevant for Latin American countries considering similar changes to their higher-education systems nowadays. As in Europe two decades ago, the debate focuses mainly on how many years a degree should take. This paper points to an equally important question: what qualification students receive if they leave before the end. A long programme with a recognised intermediate exit can widen access and completion without reducing the training required for the full degree.

A Institutional Setting: Details

B The Bologna Reform in Italy

Italy introduced the Bologna structure earlier than Portugal. Ministerial Decree 509 of November 1999 defined the new degrees, and all universities adopted them from the 2001/02 academic year. Before the reform, Italian universities ran two parallel tracks: a short *diploma universitario* of two to three years, and a *laurea* of four to six years. There was no master's tier above the *laurea*. The reform replaced both tracks with a three-year *laurea* followed by a two-year *laurea specialistica*, later renamed *laurea magistrale*. This structure is commonly known as the "3+2".

Restructuring by field. Because Italy had no master's tier before the reform, the same template had different effects than in Portugal. In engineering and computing, the five-year *laurea* became a 3+2, as in Portugal. In economics, the sciences, the humanities and the social sciences, the four-year *laurea* also became a 3+2. In Portugal, the new three-year degree in these fields replaced the old four-year one, so the first degree became shorter. In Italy, the old four-year *laurea* was instead treated as equivalent to the new five-year *laurea specialistica*, not to the new three-year *laurea*. The three-year degree was therefore a new intermediate credential in these fields too. Law initially followed the same pattern, but returned to a five-year single cycle from 2006/07. In nursing, the three-year *diploma universitario* was renamed a *laurea*, and a two-year master's became available on top. Medicine kept its six-year single cycle. Figure 11 summarises the changes. Overall, while Portugal combined two types of restructuring, Italy mostly introduced an intermediate credential.

B.1 Timing of adoption

This subsection adds detail to Figure 2. In Portugal, Decree-Law 74/2006 (March 2006) let programmes convert from 2006/07 and required conversion by 2008/09. In Italy, Ministerial Decree 509/1999 took effect once the first-level degree classes were published in October 2000, with 18 months to convert. In practice, Bondonio and Berton (2018) argue that the timing of adoption was close to random, because it depended on

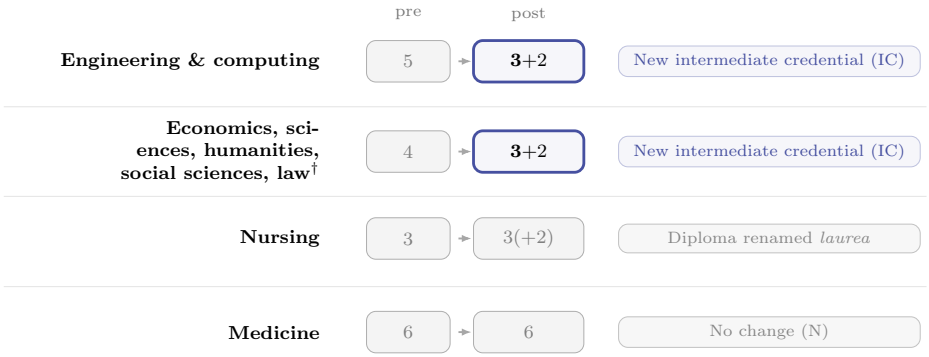
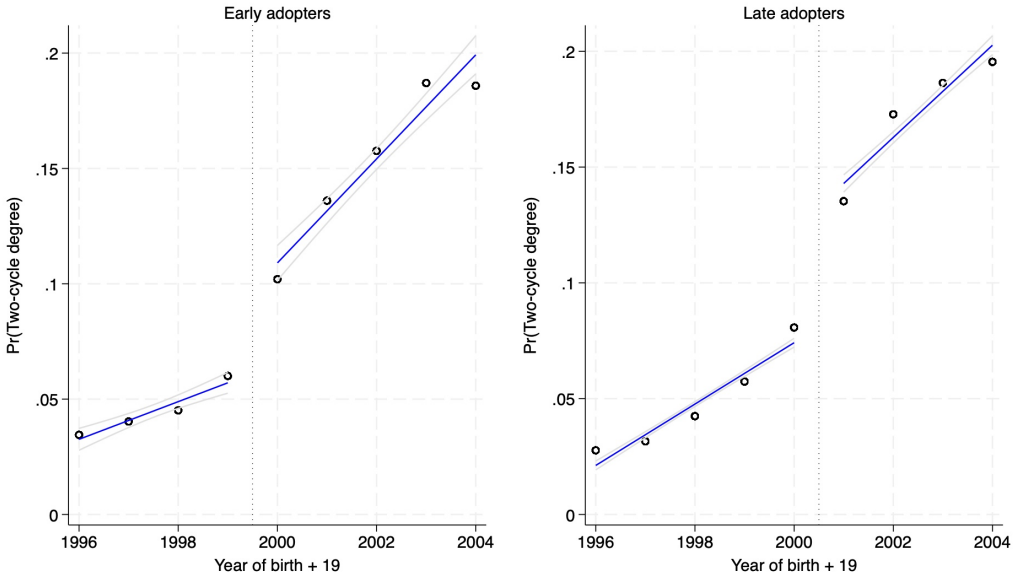


Figure 11: Years to the first university credential in Italy before and after the reform, by field

Notes: Statutory years to the first degree. (+2): master's available after the reform. Blue: new intermediate credential; grey: no change. †Law returned to a five-year single cycle from 2006/07. Other *diploma universitario* programmes not shown.

when each faculty council first met after the decree. I split Italian respondents into two groups: those living in a region where universities adopted the reform in 2000, and those living in a region where it was adopted in 2001 (Figure 12). The survey does not record where each respondent studied, so I assign each respondent the adoption year of their region of residence. This is a close proxy, because most Italian students enrol at a university in the region where they live. In 2007, 82 percent of first-year students did so (De Angelis et al., 2017).

Figure 12: Staggered adoption of the reform in Italy



Notes: Italian Labour Force Survey, 2009–2020. Share of respondents whose highest degree is a two-cycle degree, by the year they turned 19, with linear fits on each side of the adoption year. Early adopters: Piemonte, Valle d'Aosta, Friuli-Venezia Giulia and Sardinia, where universities adopted the reform in 2000/01; late adopters: all other regions (2001/02). Regions are assigned by region of residence.

B.2 What the three-year degree is worth

An earlier credential only matters if employers value it. In Portugal they were in effect obliged to, because the new degree kept the old name. The law never distinguishes *licenciaturas* by length or date: entry to the graduate career of the civil service requires "a licenciatura or higher degree" (Law 12-A/2008, art. 44), and sectoral collective agreements tie wage floors to the terms "licenciado" and "bacharel" without mentioning Bologna. By default a three-year licenciado is therefore placed on the grade a four-year licenciado used to occupy. Attempts to separate the two failed: the Constitutional Court struck down the bar association's entrance exam for post-Bologna law graduates (Acórdão 3/2011), and labour courts ruled against an employer that hired only pre-Bologna *licenciados*. In regulated professions, the value of the exit differs by field. Three-year engineering graduates can register as technical or "junior" engineers, and accountancy is open to any suitable licenciatura; the three-year degree carries no licence in medicine, psychology or school teaching, where a master's became compulsory.

Italy gave the new degree a lower but still defined place. Decree 328/2001 split the professional registers of engineers, architects, biologists, chemists, geologists and psychologists into a section for master's graduates and a "junior" section open to three-year graduates, and the three-year degree gives access to graduate grades in local government. Unlike in Portugal, however, the old four-to-six-year laurea was formally equated with the new master's, not with the three-year degree, for public-sector competitions. The pooling of old and new credentials that operates in Portugal is therefore weaker in Italy.

B.3 Other changes at the same time

Conversion of enrolled students. In both countries students already enrolled could convert to the new degrees, with credit for the exams they had passed. Cohorts just before each cutoff could therefore still leave with the new credential, so estimates for completion and later outcomes are, if anything, understated.

Entry caps and nursing (Italy). Law 264/1999 capped entry nationally to medicine, veterinary medicine, dentistry, architecture, the health professions and primary-teacher education, so entry in those fields reflects the number of places set each year. Nursing entry grew throughout 1997–2010 as training moved from regional schools into universities.

C Data: Details

C.1 Variable definitions

Educational attainment.—The main education outcome is an indicator for holding a university degree as the highest completed level. In QP this covers the *bacharelato*, *licenciatura*, *mestrado* and doctorate; once a worker is observed with a degree, the indicator stays on in later years. In QP I read it, like every education outcome, on the worker's record at age 28. In the Portuguese surveys it covers the corresponding higher-education codes, and in Italy any tertiary title. In QP, the split between *licenciatura* and *mestrado* is only available in the data for some years, hence I refrain from using this distinction and use year of education proxies where possible. However, the Portuguese LFS contains a variable that allows me to split *licenciaturas* into pre and post-reform. I use this variable when studying five-year completion of engineering degrees. The Italian LFS contains a similar vaiable, allowing me to distinguish single from two-cycle degrees, which I use to measure the take-up of the new structure, and distinguish when a student studied for four or more years (a single-cycle degree or a *laurea specialistica*).

Years of university.—In the surveys, years of university equal the age at which the respondent left education minus 18 (19 in Italy) for graduates, and zero for everyone else. QP has no such question. I proxy it with the age at which a worker is first observed with their highest degree, minus 18, for workers who hold a degree at age 28, and zero for everyone else; because this also counts any delay before the first private-sector job, I validate it against the surveys.

Field of study.—I group fields by what the reform did to the length of the first credential (Section 2). In QP, the last two digits of the education code give the field of the highest degree, in the two-digit fields of ISCED 1997. The first group contains the fields where a five-year degree gained a three-year exit ("5 to 3+2"): the sciences, mathematics, computing and engineering (ISCED 42–58). The second contains the fields where a four-year first degree became three years ("4+2 to 3+2"): business and administration (34), education (14), and the humanities and the arts (21 and 22), social sciences and journalism (31 and 32) and the service fields (76 and 81–86). The third contains the fields with no usable early exit ("no change"): health (72), together with veterinary science (62 and 64), and law (38). For some graduates the field is

missing. The Portuguese surveys record only the broad fields of the national classification (CNAEF). In these, the social sciences, business and law share one code, which I place in the 4+2 to 3+2 group, so the no-change group contains health only. In Italy I group degree subjects into STEM, humanities and teaching, the social sciences (including law and psychology), and health.

Occupation and sector.—The occupation describes what a worker does, and the sector describes what the employer produces. An engineer employed by a bank, for example, works in an engineering occupation but in the finance sector. QP records a four-digit occupation code for each worker and a sector code for each employer. Occupations follow CPP-2010, the Portuguese version of ISCO-08, from 2010 onwards, and CNP-94 before. Sectors follow CAE Rev. 3, the Portuguese version of NACE Rev. 2, from 2007 onwards. I measure both on the worker's record at age 28. For every cohort I study, that record falls in 2010 or later, so each outcome uses a single classification and no mapping between classifications is needed. I use occupations at the three-digit level and sectors at the two-digit level. For each group of fields, I define the occupations and the sectors that match it. Engineering matches engineering and ICT professionals and technicians (ISCO-08 214, 215, 251, 252 and 311) and IT and engineering firms (CAE 62, 63 and 71). Business, education, the humanities and the arts match business, finance, teaching and cultural professionals (ISCO-08 231–235, 241–243, 262, 264 and 265) and firms in finance, legal and accounting services, management consultancy, education, the arts and culture (CAE 64–66, 69, 70, 85, 90 and 91). Health matches health professionals and associate professionals (ISCO-08 221–226 and 321–325) and health care providers (CAE 86). Each outcome is an indicator for working in the matching occupation or sector. It is defined for every worker, and a worker with a missing code counts as outside it. Because QP does not cover civil servants, the education sector includes only private schools and universities. Public hospitals are included, because most of them report to QP as public enterprises.

Wages and earnings.— The QP hourly wage is monthly October pay divided by normal monthly hours, deflated to 2010 euros with the consumer price index. Earnings and hours are each capped at their 99th percentile within year before the ratio is taken. The wage outcome is one value per worker, and it is measured relative to older workers. For each calendar year t, the reference wage $\overline{\log w_t}^{40\text{–}45}$ is the average log real hourly wage of workers aged 40 to 45 in that year; all of them turned 18 before the reform. For each year in which

a worker is aged 28 to 30, I take the gap between the worker's log hourly wage and the reference wage of that year, and I average these gaps over the worker's years at ages 28 to 30 (equation (12)). Appendix D.4 explains why the comparison with older workers removes calendar-time effects.

I measure cumulative real earnings to a given age by summing annual earnings, set to zero in years the worker is not in QP. I pad each worker's record to one row per age from school-leaving age to age A, set annual earnings to zero where no earnings are observed, and sum. Zeros capture delayed entry; the measure does not condition on employment.

In the surveys employment follows the ILO definition and pay is set to zero for those not employed.

Enrolment and dropout.—From IEFA I define entry to higher education as having completed a degree or having started one and left, and dropout as having started and left without a degree. I report dropout as a share of the whole cohort.

C.2 Age at measurement

Figure 13 justifies the use of age 28 for the analysis. The share of people holding a university degree increases with age, so measuring outcomes at earlier ages such as age 24 would miss some individuals who are still studying and have not yet entered work. This increasing trend stabilizes after age 27.

D Empirical Strategy: Details

D.1 What the estimates measure

The coefficient τ measures the effect of facing the reformed degree structure at the age of entry, averaged over the whole cohort. The average includes people who never enrol, because they are part of the population whose choices the reform could change. I do not rescale τ by a first stage. The reform did not offer a treatment that some students took up and others refused; it changed the options open to every student, both the decision to enrol and the decision of when to stop.

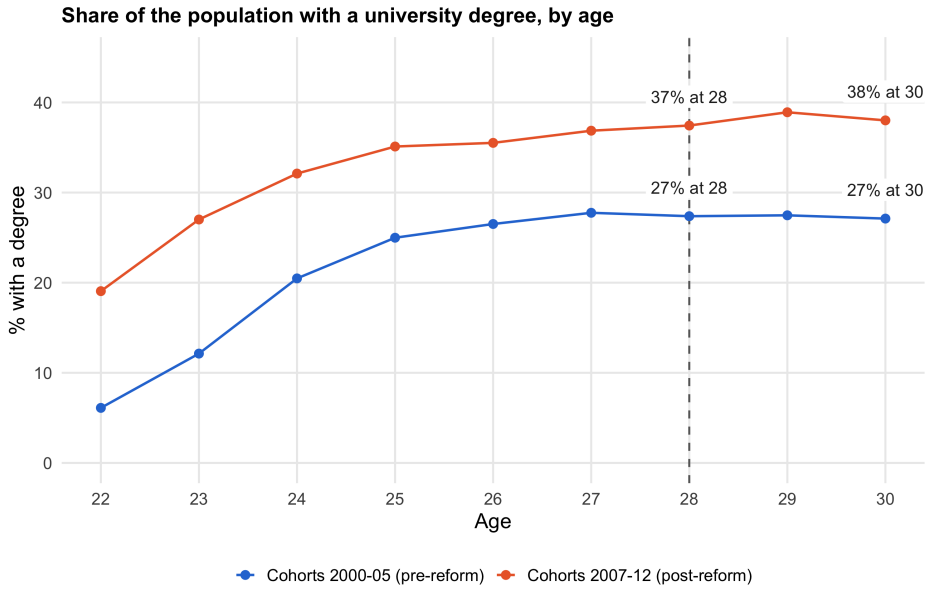


Figure 13: Share of the population with a university degree, by age, Portugal

Notes: Portuguese Labour Force Survey, 2004–2024. Share of everyone in the cohort whose highest completed level is a university degree (*bacharelato*, *licenciatura*, *mestrado* or doctorate), by age. Blue: the six cohorts that turned 18 in 2000–2005, before the reform. Orange: the six cohorts that turned 18 in 2007–2012, after it. Every cohort of each group is observed at every age shown. The dashed line marks age 28, the age at which education is measured.

D.2 Estimation and standard errors

I estimate equation (2) on cohort means. Each worker has one value of each outcome: education at age 28, and the wage relative to workers aged 40 to 45, averaged over ages 28 to 30 (equation (12)). I average these values within each cohort, so each worker counts once, separately for women and men and for everyone together. I then fit a line on each side of the cutoff to the twelve cohort means by ordinary least squares. The standard errors are the conventional standard errors of this regression, and p-values come from a t distribution with eight degrees of freedom (twelve cohorts minus four parameters). The cohort is therefore the unit of inference. This follows Donald and Lang (2007): with few groups, cluster-robust standard errors computed on individual observations are biased downwards, whereas a regression on group means, with a t distribution whose degrees of freedom are the number of groups minus the number of parameters, gives valid inference when the cohort-level errors are close to normal. This allows for shocks common to everyone in a cohort, such as the labour market they entered, which standard errors computed on individual workers would ignore. Each cohort mean rests on tens of thousands of workers, so the sampling error of the cohort means is small next to the variation across cohorts, and I do not add it. To test whether the effect differs between

women and men, I compare the two estimates with a z-test that treats the two samples as independent. The Italian regressions are run on individual respondents, with heteroskedasticity-robust standard errors.

D.3 Validity of the design

Density of the running variable Birth year cannot be manipulated, but the number of workers per cohort in *Quadros de Pessoal* could still jump at the cutoff if the reform changed who enters the private sector. I test for a break in the density of the running variable in the spirit of McCrary (2008). For each cohort bin j I compute $\hat{f}(z_j) = N_j/(Nh)$, where N_j is the number of workers in the bin, N the total and $h = 1$ the bin width, and estimate

$$\log \hat{f}(z_j) = \gamma_L + \delta_L z_j + u_j \quad (z_j < 0), \qquad \log \hat{f}(z_j) = \gamma_R + \delta_R z_j + v_j \quad (z_j \geq 0), \tag{11}$$

weighting each bin by N_j and using a triangular kernel within a symmetric window around the cutoff. The discontinuity in the log density is $\hat{\gamma}_R - \hat{\gamma}_L$. Figure 14 shows no visible break. The estimated jump in the log density is 0.019, with a standard error of 0.020.

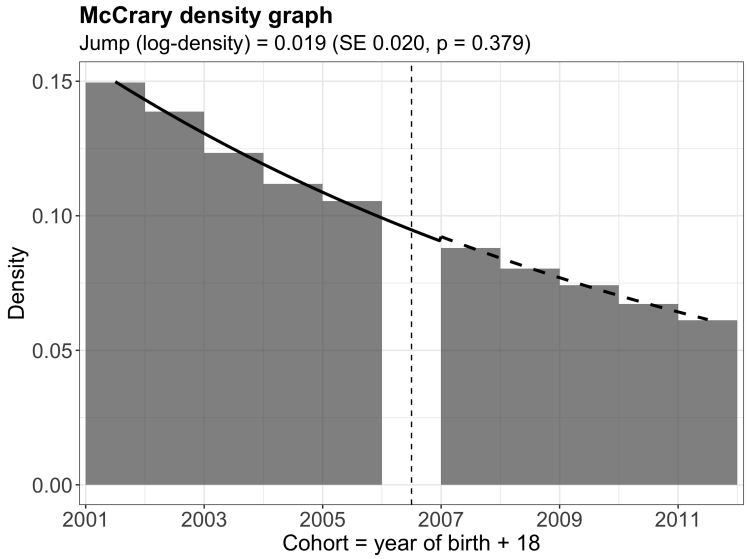


Figure 14: Density of the running variable in *Quadros de Pessoal*

Notes: Quadros de Pessoal. Share of workers by the year they turned 18, with fits from equation (11) on each side of the cutoff; the dotted line marks the excluded 2006 cohort. The estimate at the top of the panel is the jump in the log density at the cutoff.

No delayed entry Students who turned 18 shortly before 2007, and who expected the reform, could have postponed entry to enrol in the shorter degrees. Such students would then be counted as untreated although

they studied under the new structure, and the pre-reform cohorts closest to the cutoff would be selected. Postponing entry usually means working in the meantime, so I estimate equation (2) with an indicator for being observed in employment between leaving secondary school and completing a university degree. Table 7 shows that the discontinuity is close to zero and precisely estimated for women and men alike.

Table 7: Employment between secondary school and university

	Whole sample	Women	Men
RD jump	0.001	0.002	−0.000
	(0.001)	(0.002)	(0.002)
Baseline	0.105	0.128	0.085
p-value, women = men		0.283	

Notes: Quadros de Pessoal. The outcome is an indicator for being observed in employment between leaving secondary school and completing a university degree. Estimates of equation (2). Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

D.4 Wages relative to older workers in the same year

I measure each wage relative to the wages of workers aged 40 to 45 in the same calendar year (Section 4). Let w_{it} be the hourly wage of worker i in year t, and let $\overline{\log w_t}^{40\text{–}45}$ be the average log hourly wage of workers aged 40 to 45 in year t. The wage outcome is

$$Y_i = \frac{1}{|T_i|} \sum_{t \in T_i} \left(\log w_{it} - \overline{\log w_t}^{40\text{–}45} \right), \tag{12}$$

where T_i is the set of years in which worker i is observed with a wage at ages 28 to 30.

A person who turns 18 in year c is aged a in calendar year $t = c + a - 18$. The wage outcome uses ages 28 to 30, so each cohort is paid in its own three calendar years. Figure 15 shows these years for each cohort in the blue bars. The 2000 cohort is paid in 2010 to 2012, and the 2012 cohort in 2022 and 2023; the data end in 2023, so this last cohort is observed at ages 28 and 29 only. Because the ages are fixed, the average calendar year rises one for one with the cohort. A comparison across cohorts is therefore also a comparison across calendar years. The pre-reform cohorts are paid between 2010 and 2017, mostly during the crisis. The post-reform cohorts are paid between 2017 and 2023, during the recovery. Without an adjustment, these changes in pay would show up in the cohort trends and in the jump at the cutoff.

Calendar-year effects estimated on the young workers themselves would separate cohorts from years only

through the years in which cohorts from both sides of the cutoff are paid. At ages 28 to 30 there is a single such year, 2017, when the 2005 cohort is 30 and the 2007 cohort is 28. The estimate would then rest on one year. I instead measure calendar time on a group of workers outside the estimation sample.

The reference group is workers aged 40 to 45. For each calendar year, I compute their average log hourly wage, and I measure the wage of every young worker paid that year relative to it. The orange bars in Figure 15 show that these older workers are paid in exactly the same years as the estimation sample, 2010 to 2023, and always at the same ages. The dotted line marks one year, 2017. That year, the 2005 and 2007 cohorts are paid at ages 30 and 28, and the workers who turned 18 between 1990 and 1995 are paid at ages 40 to 45. Whatever changes pay in 2017, such as the state of the economy, the minimum wage or inflation, reaches both groups. The older workers all turned 18 between 1983 and 2001, so the reform did not change their education. I also hold the mix of workers in the reference group fixed over time, so that its average moves with pay and not with who is employed.

To see what the comparison removes, write the log hourly wage of a young worker i in year t as $\log w_{it} = \theta_i + \mu_t + e_{it}$, where θ_i is the worker's own pay level and μ_t is the change in pay that year common to all workers. Write the average log wage of the reference group as $\overline{\log w_t}^{40\text{–}45} = \kappa + \mu_t + v_t$, where κ is the fixed pay level of the reference group. The gap between the two is

$$\log w_{it} - \overline{\log w_t}^{40\text{–}45} = \theta_i - \kappa + e_{it} - v_t, \tag{13}$$

in which the calendar term μ_t no longer appears. The reform can change θ_i for the post-reform cohorts, through their education, field and employer, but it cannot change κ. The jump in the cohort means of the gap therefore measures the effect of the reform on the wages of the young cohorts.

The comparison assumes that a shock in a given year moves the log wages of workers aged 28 to 30 and of workers aged 40 to 45 by the same amount. It would fail if, for example, the crisis cut the pay of young workers by more than the pay of older workers. The pre-reform cohorts allow a check on this. They are paid from 2010 to 2017, through the crisis and the start of the recovery. If those years had moved young and older workers' pay differently, the relative wage would trend from one pre-reform cohort to the next.

For each worker, I average the gaps over the years in which they are observed with a wage at ages 28 to 30, as in equation (12). As in Mountjoy (2022), a worker enters the sample if they are observed at least once

at these ages. The average age of the workers is close to 29 in every cohort, and 28.5 in the 2012 cohort. Mountjoy (2022) averages earnings over the same ages without a calendar adjustment. He does not need one, because his comparison is between students of the same cohort, who are paid in the same years. The comparison here is across cohorts, so calendar time has to be removed.

Card and Cardoso (2012) face the same problem with the same data. They compare the wages of birth cohorts in which different shares of men were conscripted. As here, no two of their cohorts are observed at the same age in the same year. They separate cohort, age and year effects by assuming that, within each year, log wages follow a smooth function of age. Because every cohort here is observed at the same ages, the comparison with older workers needs no assumption on how pay varies with age.

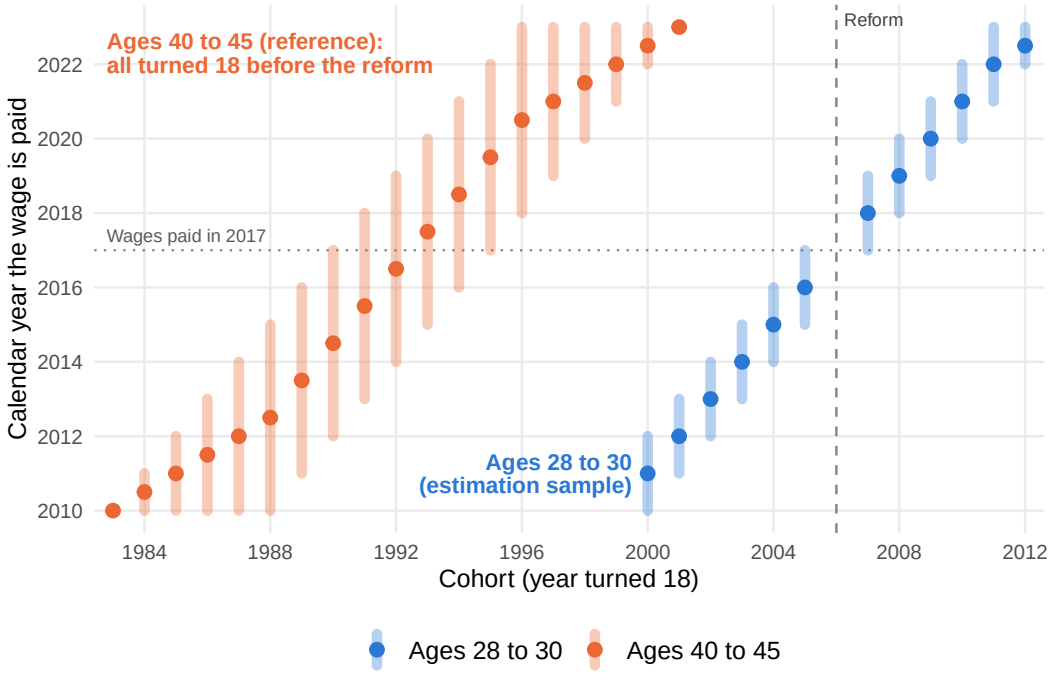


Figure 15: Calendar years in which each cohort's wage is measured

Notes: Each bar shows the calendar years, within 2010–2023, in which a cohort is observed at the stated ages; the point marks the average year. A person who turns 18 in year c is aged a in year $c + a - 18$. Blue: cohorts 2000–2012 at ages 28 to 30, the estimation sample; the 2006 cohort is excluded. Orange: cohorts at ages 40 to 45, the reference group, all of whom turned 18 between 1983 and 2001. The dotted line marks 2017, and the dashed line the reform.

D.5 Wage effects at other ages, with other reference groups and in narrower windows

Table 1 measures the wage over ages 28 to 30, relative to workers aged 40 to 45 in the same year, with six cohorts on each side of the cutoff. Table 8 changes each of these choices in turn. Averaging the wage over ages 28 to 29 gives 2.1 percent, and over ages 29 to 30, 2.4 percent (columns 2 and 3). Older ages are not possible, because the data end in 2023 and the youngest post-reform cohorts would drop out. Measuring the wage relative to workers aged 35 to 45, or 35 to 40, gives 1.8 and 1.7 percent (columns 4 and 5). Keeping five cohorts on each side gives 1.8 percent, and keeping four gives 1.3 percent (columns 6 and 7). In every column the effect is larger for women; the difference between women and men is significant at the 5 percent level only with four cohorts on each side.

Table 8: Effect of the reform on log hourly wages: other ages, reference groups and windows

		Ages		Reference group		Cohorts on each side	
	Main	28–29	29–30	35–45	35–40	5	4
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
All	0.021***	0.021***	0.024***	0.018**	0.017**	0.018***	0.013**
	(0.005)	(0.006)	(0.005)	(0.006)	(0.006)	(0.005)	(0.003)
Women	0.025***	0.024**	0.028***	0.023***	0.022**	0.023***	0.018**
	(0.006)	(0.007)	(0.006)	(0.007)	(0.007)	(0.006)	(0.005)
Men	0.015**	0.015**	0.019***	0.012**	0.011*	0.012**	0.007**
	(0.004)	(0.005)	(0.004)	(0.005)	(0.005)	(0.004)	(0.002)
p-value, women = men	0.18	0.33	0.22	0.19	0.22	0.14	0.05
Cohorts, pre / post	6 / 6	6 / 6	6 / 6	6 / 6	6 / 6	5 / 5	4 / 4
Workers	1,017,067	924,454	929,084	1,017,067	1,017,067	841,183	660,340

Notes: Quadros de Pessoal, one observation per worker, as in Table 1, column 1. Column 1 is that estimate. Columns 2–3: the worker's mean over ages 28–29 or 29–30 only. Columns 4–5: the reference group is workers aged 35–45 or 35–40 who turned 18 before 2006, instead of 40–45. Columns 6–7: five or four cohorts on each side of the excluded 2006 cohort (2001–2005 and 2007–2011; 2002–2005 and 2007–2010). Each column fits equation (2) to the cohort means; standard errors from that regression, p-values from a t distribution with the number of cohorts minus four degrees of freedom; women = men: z-test. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.6 AKM decomposition of wages

A wage gain can arise because workers from reformed cohorts are paid more in the same firms or because they work in firms that pay all their workers more. To separate the two I estimate a two-way fixed-effects wage equation (Abowd et al., 1999),

$$\log w_{it} = \theta_i + \psi_{J(i,t)} + \mu_t + h(a_{it}) + \epsilon_{it}, \tag{14}$$

where θ_i is a worker effect, $\psi_{J(i,t)}$ the pay premium of the firm $J(i,t)$ that employs worker i in year t, μ_t a year effect and $h(\cdot)$ a quadratic in age. Worker and firm effects are separately identified only among firms linked by workers who move between them, so I estimate equation (14) on the largest connected set of firms and workers.

I then run the discontinuity regression on the wage and on its firm component. I measure the firm component as I measure the wage, relative to workers aged 40 to 45 in the same year. Let $\overline{\psi}_t^{40\text{–}45}$ be the mean firm effect of the employers of workers aged 40 to 45 in year t, weighted as the reference wage in equation (12). For each worker, let

$$\bar{\psi}_i = \frac{1}{|T_i|} \sum_{t \in T_i} \left(\hat{\psi}_{J(i,t)} - \overline{\psi}_t^{40\text{–}45} \right), \tag{15}$$

with the same years T_i as in equation (12). Because $Y_i = \bar{\psi}_i + (Y_i - \bar{\psi}_i)$, and because equation (2) is linear and is estimated on the same workers for each term, the wage jump splits exactly into two parts:

$$\Delta^w = \Delta^\psi + \Delta^{w-\psi}, \qquad \text{firm share} = \frac{\Delta^\psi}{\Delta^w}. \tag{16}$$

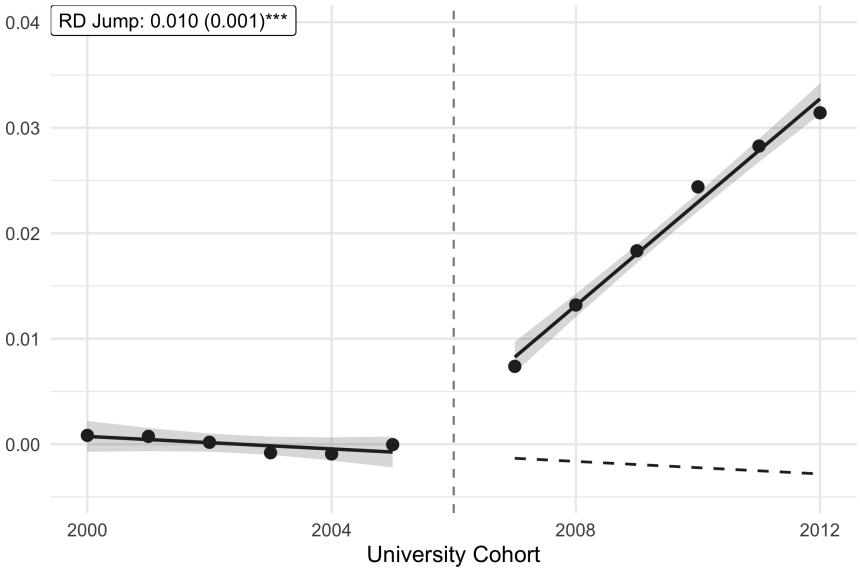


Figure 16: Firm effect relative to workers aged 40–45, by cohort

Notes: Cohort means of $\bar{\psi}_i$ in equation (15), relative to the pre-reform mean, with the two lines of equation (2), 95 percent confidence bands and the pre-reform line extended. The jump is Table 1, column 2. See the notes to Figure 3.

The firm component Δ^ψ measures how much of the wage jump comes from working in firms with higher

pay premia than the firms of older workers in the same years. The remainder $\Delta^{w-\psi}$ contains the worker effects and everything else. Firm premia are held fixed over the sample period, so the firm share captures moves towards firms that pay more to all their workers, not changes in how firms pay. The reference wage contains the firm effects of the older workers' employers, so the wage and the firm effect are now measured against the same people, and a change over time in the firms where older workers are employed is removed from both. Equation (14) is estimated on every worker-year of the cohorts that turned 18 in 2000–2012. Figure 16 shows the RD on $\bar{\psi}_i$.

E Results by sex

This appendix shows, for women and men separately, the figures whose pooled versions are in Section 5. The paragraphs on women and men in Section 5 discuss them.

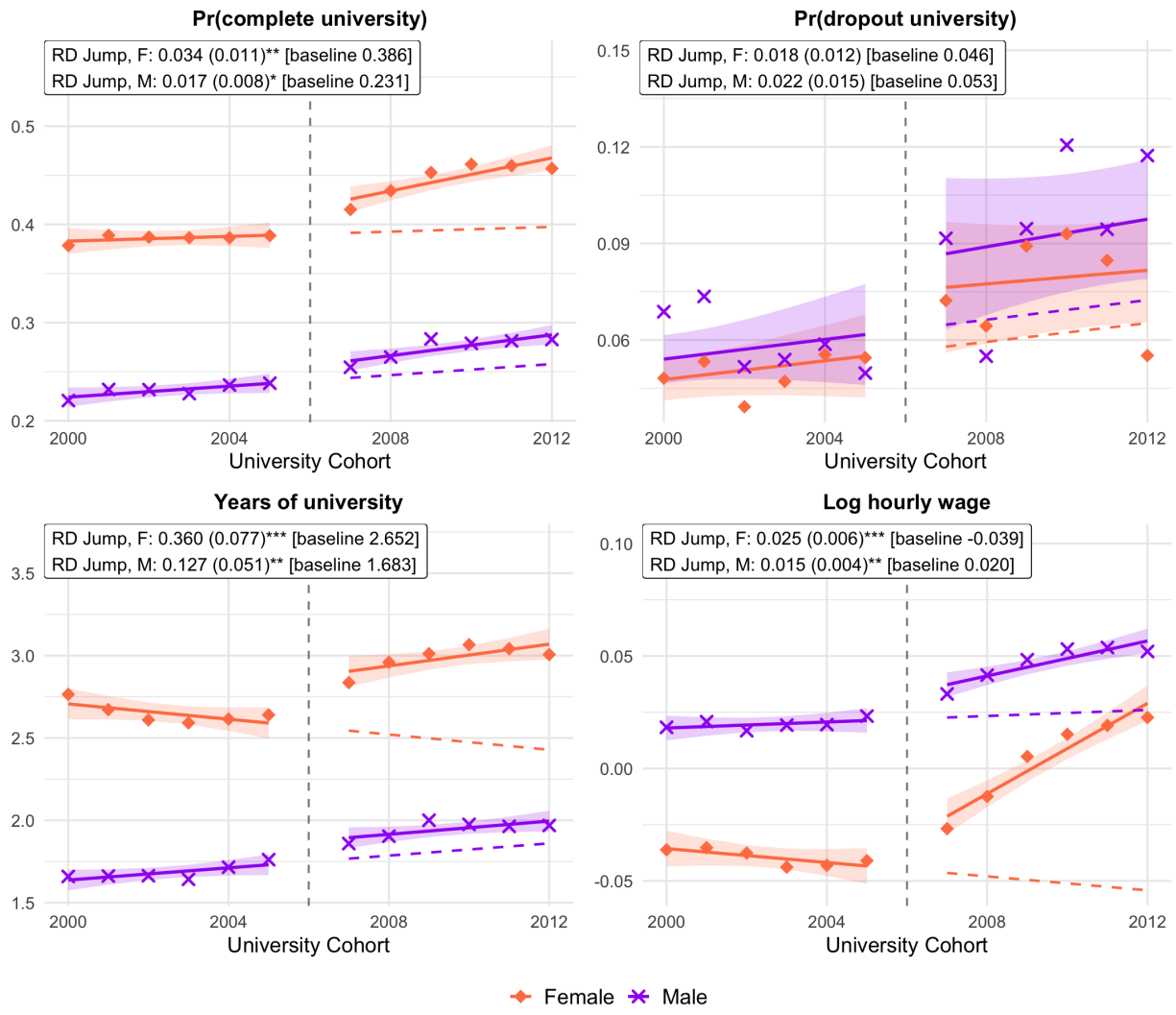


Figure 17: University completion, dropout, years of university and hourly wages, women and men

Notes: As Figure 3, for women and men separately (IEFA dropout sample: 7,505 women and 7,044 men; QP sample: 489,387 women and 527,680 men). Each panel has its own y-axis range. Boxes report the estimate of τ for each sex, its standard error and that sex's pre-reform mean. Completion in other data sources: Table 12.

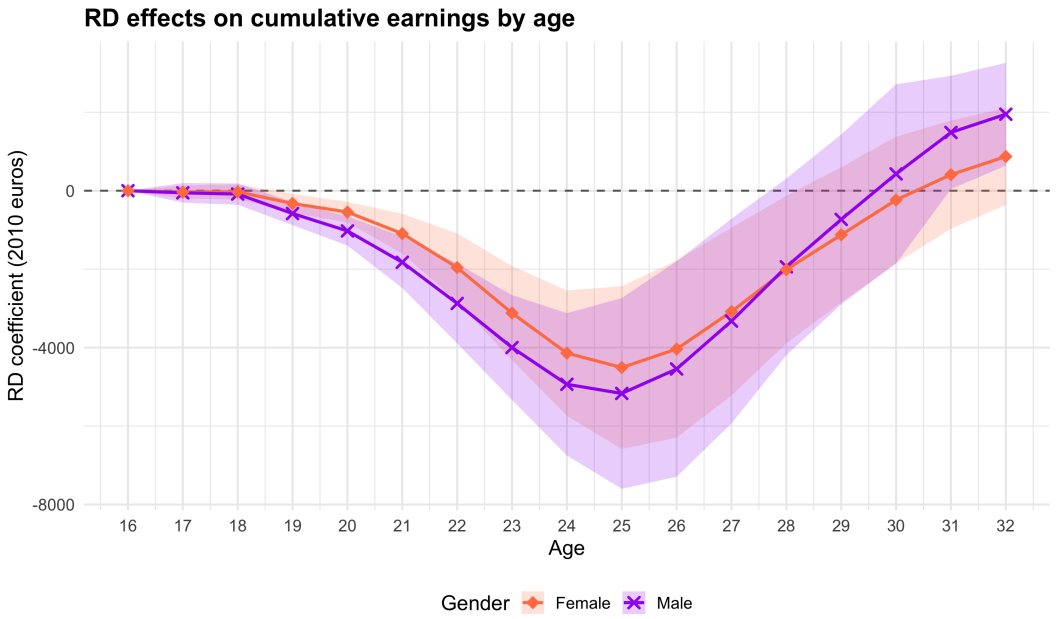


Figure 18: Effect of the reform on cumulative earnings by age, women and men

Notes: As Figure 4, for women and men separately.

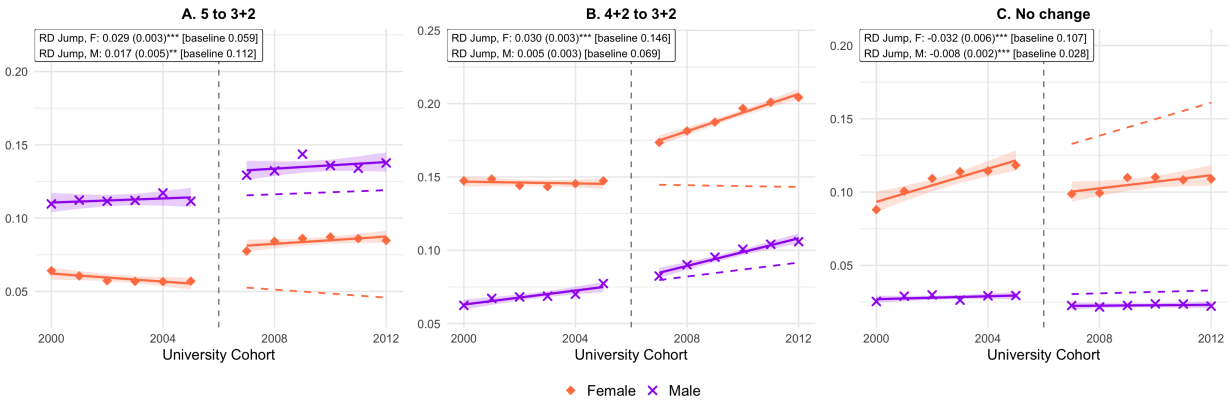


Figure 19: University degrees by what the reform did to the field, women and men

Notes: As Figure 5, for women and men separately.

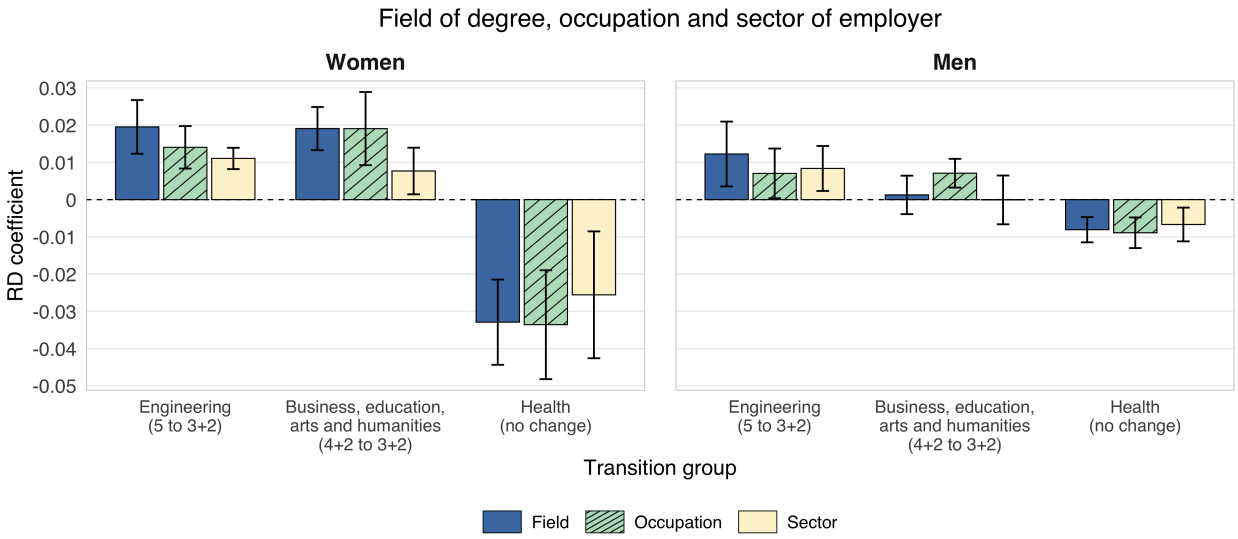


Figure 20: Field of degree, occupation and sector of employer, women and men

Notes: As Figure 6, for women and men separately.

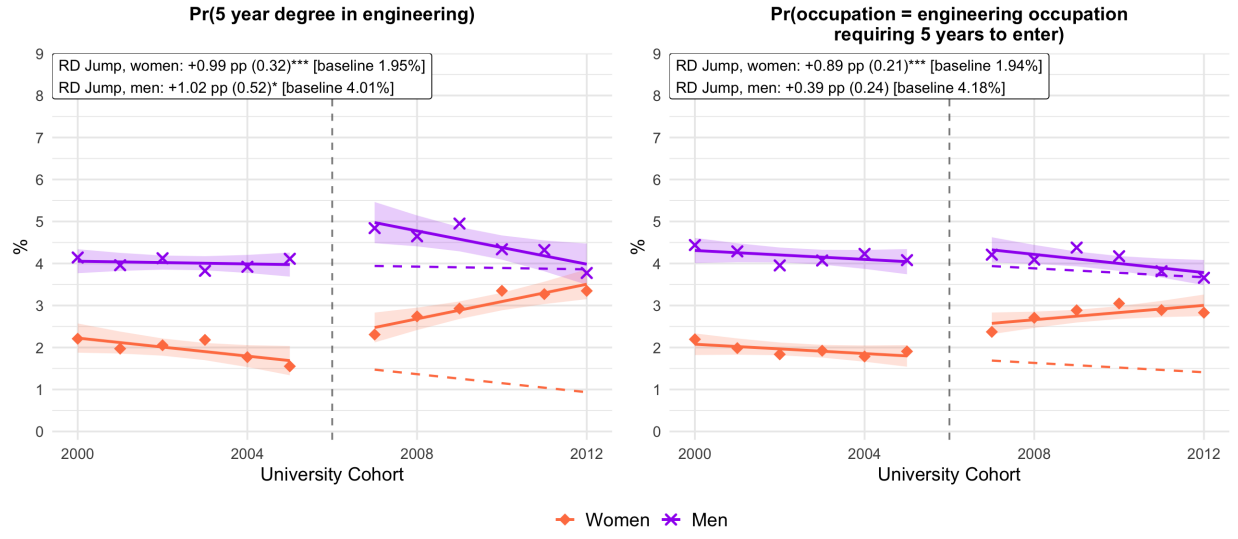


Figure 21: Five-year engineering degrees and five-year engineering occupations, women and men

Notes: As Figure 7, for women and men separately.

F Where the wage gain accrues: firm types

Section 5.1 shows that post-reform cohorts earn 2.1 percent more at ages 28 to 30, and that a little under half of this gain comes from working at firms that pay all their workers more (Table 1). The firm effect does not say in which firms the gain is realised. This appendix therefore splits the gain by the type of firm people work in. The split adds up exactly to the wage effect, and each of its pieces is an RD on an outcome defined for every worker, so no piece conditions on where the worker ends up.

Firm types. I classify each employer by its three-digit CAE Rev. 3 sector. QP codes some firms at division level in some years and at group level in others (banks move between 641 and 640, insurers between 651 and 650), so for those divisions I use the division code in every year. I group sectors into three types. *IT and engineering*: computer programming and IT consulting (division 62), information services such as data processing and hosting (63), and architecture and engineering firms (71), the same definition as in Figure 6. *Business and finance*: banking, insurance and finance (divisions 64–66), head offices (701), management consultancy (702), accounting, audit and tax (692), and other professional and technical services (749). *All other firms*: everything else, including rental, staffing, travel, security, cleaning and office support (divisions 77–82).

Outcomes. Let $D_{it} = \log w_{it} - \overline{\log w_t}^{40\text{–}45}$ be the wage gap of equation (12), so that Y_i is its mean over the years T_i. Let $k(i,t)$ be the type of the firm that employs worker i in year t, and let c be the mean of Y_i in the pre-reform cohorts. For each type k I define

$$y_{ik} = \frac{1}{|T_i|} \sum_{t \in T_i} (D_{it} - c)\, \mathbf{1}\{k(i,t) = k\}, \tag{17}$$

which is zero for a worker never employed in a type-k firm at these ages. Summing over types gives $\sum_k y_{ik} = Y_i - c$. Equation (2) is linear in the outcome, it is estimated on the same workers for each type, and c is a constant, so the RD estimates on the y_k add up exactly to the RD on Y. The type of firm is read in each year, not at age 28 as in Section 5.2, so that the split adds up to the wage over ages 28 to 30.

The mean of y_k in a cohort is $s_k(\mu_k - c)$, where s_k is the average share of workers' years spent in type k

and μ_k is the average wage gap in those years. Its jump at the cutoff is

$$\Delta\big[s_k(\mu_k - c)\big] = \underbrace{\Delta s_k\,(\mu_k^- - c)}_{\text{more workers in } k} + \underbrace{s_k^-\,\Delta\mu_k}_{\text{higher pay inside } k} + \underbrace{\Delta s_k\,\Delta\mu_k}_{\text{interaction}}, \tag{18}$$

where a superscript $-$ denotes the limit from the pre-reform side. A firm type therefore adds to the wage effect if post-reform cohorts are paid more inside it, if more of them work there and it paid above the pre-reform average, or if fewer of them work there and it paid below that average.

The results say where the reform's wage effect is realised, not how large it would have been had workers not changed sector. Sector is itself an outcome of the reform, so a comparison of wages *within* a sector would compare post-reform workers with pre-reform workers who would have chosen differently. The outcomes y_k avoid that comparison because they are defined for everyone.

Results. Table 9 reports the RD on each y_k, and Figure 22 shows the cohort means. The three types cover every worker, so the three RDs add up exactly to the wage effect. IT and engineering firms account for 28 percent of the effect, and business and finance firms for 24 percent. These firms employ about a tenth of pre-reform workers but account for just over half of the effect. All other firms account for the remaining 48 percent, which is significant only at the 10 percent level.

The two types add to the effect in different ways. IT and engineering firms employ more post-reform workers. Their share of workers rises by 1.0 percentage point, from a baseline of 4.1 percent, and this rise accounts for more than half of their part of the effect. Business and finance firms do not: their share of workers does not change significantly, so their part comes mostly from higher pay inside these firms. This part should be read with care. It falls across the pre-reform cohorts and rises across the post-reform cohorts, and between the 2005 and 2007 cohorts it rises by less than half of the estimated jump. The fall among pre-reform cohorts reflects a decline in banking employment that stops after the cutoff. The parts do not differ significantly between women and men in any column (Figure 23).

Table 9: Hourly wages at ages 28–30: where the effect accrues, by the type of firm people work in

		Part of the effect in		
	Log wage (1)	IT and engineering (2)	Business and finance (3)	All other firms (4)
All	0.021***	0.006***	0.005***	0.010*
	(0.005)	(0.001)	(0.001)	(0.005)
Baseline	−0.010	—	—	—
Women	0.025***	0.005***	0.006***	0.014**
	(0.006)	(0.001)	(0.001)	(0.006)
Baseline	−0.039	—	—	—
Men	0.015**	0.006***	0.004***	0.004
	(0.004)	(0.001)	(0.001)	(0.004)
Baseline	0.020	—	—	—
p-value, women = men	0.18	0.65	0.26	0.16
Share of the effect, all (%)	100	28	24	48
Share of workers, pre-reform (%)	100	4.1	5.4	90.5
Cohorts, pre / post	6 / 6	6 / 6	6 / 6	6 / 6
Workers	1,017,067	1,017,067	1,017,067	1,017,067

Notes: Estimates of τ in equation (2) at the 2007 cohort (Section 4). *Quadros de Pessoal*, one observation per worker, as in Table 1, column 1. Column 1 is that estimate: the worker's mean, over the years in which they are aged 28–30, of the log real hourly wage minus the mean log hourly wage of workers aged 40–45 in the same year (equation (12)). Columns 2–4: the worker's mean, over the same years, of (that gap minus its pre-reform mean) times an indicator for working in that type of firm in that year, zero otherwise; they add up exactly to column 1. No column conditions on the type of firm. A type of firm adds to the effect when more people work there at above-average pay, or when those who work there are paid more. IT and engineering: CAE divisions 62, 63 and 71. Business and finance: 701, 702, 692, 749 and divisions 64–66. All other firms include divisions 77–82 (rental, staffing, travel, security, cleaning and office support). Share of workers: a worker employed in two types of firm counts in each in proportion to their years there. Baseline: pre-reform mean; the parts in columns 2–4 have no meaningful baseline. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

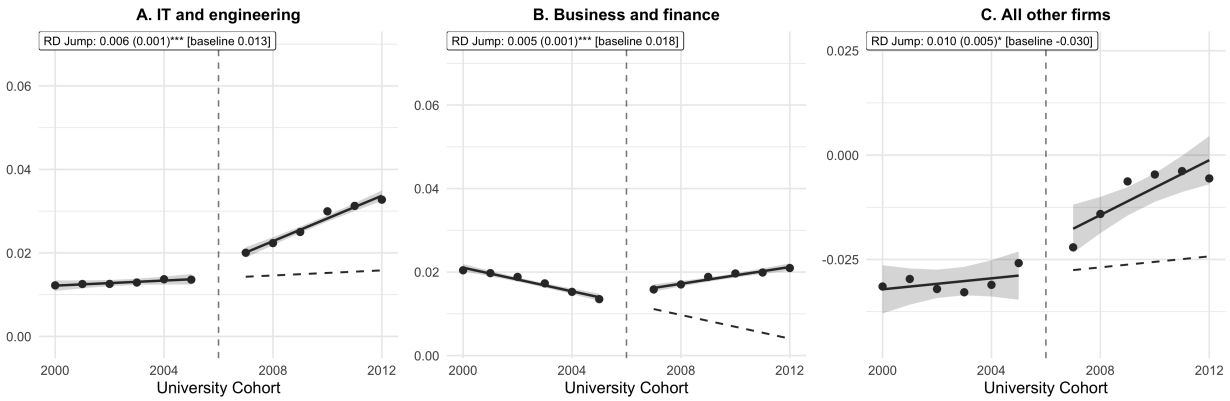


Figure 22: Where the wage effect accrues, by the type of firm, by cohort

Notes: Cohort means of each part in Table 9, columns 2–4, with the two lines of equation (2), 95 percent confidence bands and the pre-reform line extended. See the notes to Figure 3.

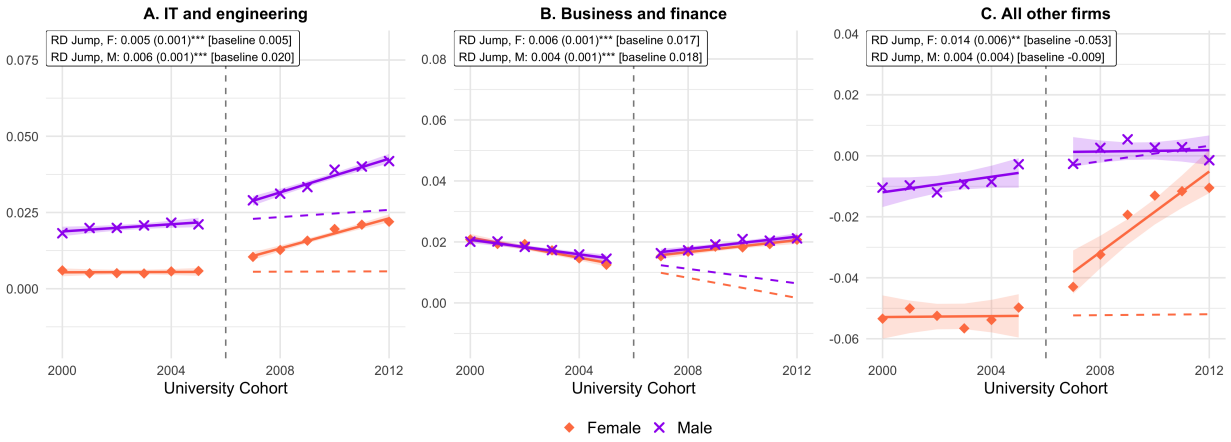


Figure 23: Where the wage effect accrues, by the type of firm, women and men

Notes: As Figure 22, for women and men separately.

G Placebo cutoffs

The main results compare cohorts on either side of 2007. If the cohort series had breaks of this size from time to time, a jump at 2007 would not be evidence of the reform. I therefore move the cutoff to cohorts where nothing happened. I place it at 2002, 2003 and 2004, using only the pre-reform cohorts 2000–2005, and at 2009, 2010 and 2011, using only the post-reform cohorts 2007–2012. I run this for the six main *Quadros de Pessoal* outcomes, each measured as in the main text, with one observation per worker: university completion and the three field groups (Figures 3 and 5), years of university (Figure 3) and the log hourly wage (Table 1, column 1).

Figure 24 and Table 10 show that the jump is specific to 2007. In the whole sample, no placebo jump is as large as the actual jump. Depending on the outcome, the largest of the six placebos is between 34 and 63 percent of the actual jump, and the largest with the same sign as the actual jump is at most 34 percent of it. Five of the 36 placebo estimates are significant at the 5 percent level, which is more than expected by chance, but four of these five have the opposite sign to the actual jump. These standard errors are not reliable, because each placebo rests on six cohort means and has two degrees of freedom. What matters is the size of the placebo jumps relative to the actual one. The placebos at 2010 and 2011 are negative for completion, years of university, the 5 to 3+2 group and the wage, because the post-reform series flattens in the latest cohorts. These placebos have the opposite sign to the actual jump, so they cannot account for it.

For women, the largest placebo is at most 59 percent of the actual jump for every outcome (Figure 25 and Table 11). For men, the actual jumps are smaller, so the ratios are larger. The largest placebo is between 48 and 85 percent of the actual jump for four of the six outcomes, and about as large as the actual jump for the other two: completion and the 4+2 to 3+2 group. In these two cases the placebo has the opposite sign, and men's jump in the 4+2 to 3+2 group is itself not significantly different from zero.

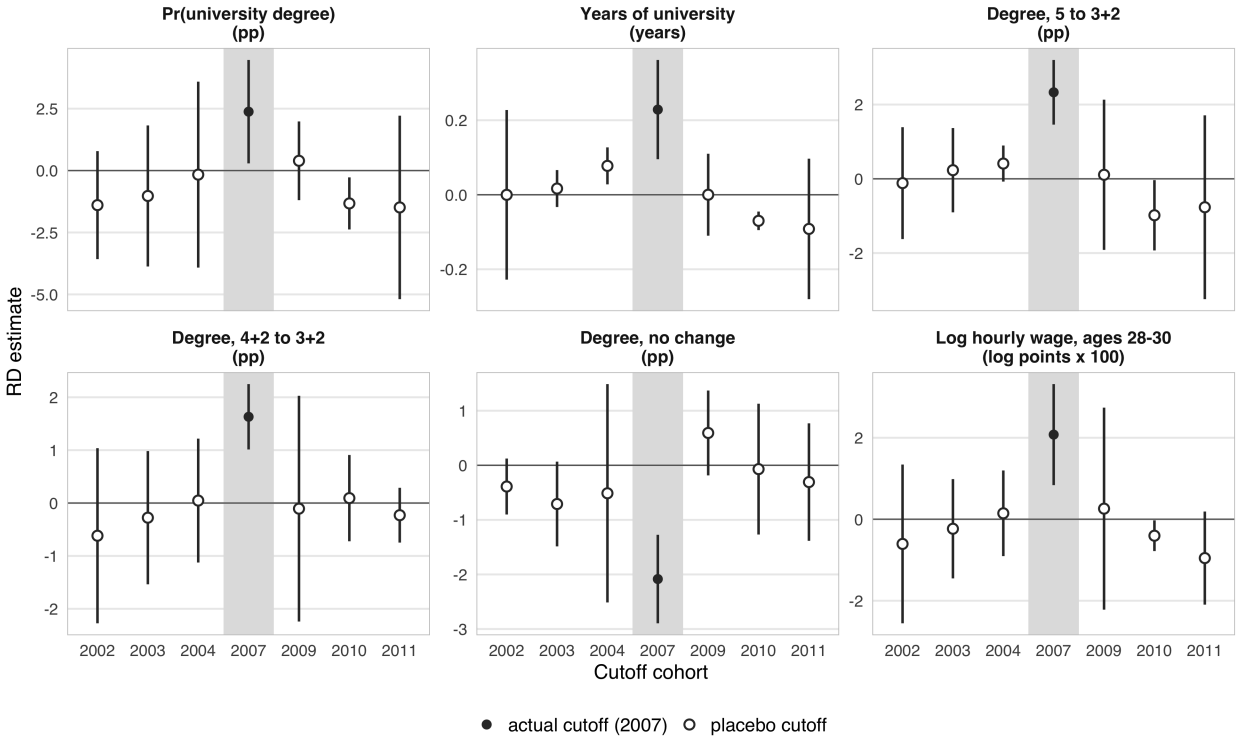


Figure 24: Jumps at the actual and at placebo cutoffs

Notes: *Quadros de Pessoal*, one observation per worker. Each point is the RD estimate at the cutoff cohort on the horizontal axis, with its 95 percent confidence interval from a t distribution with the regression's degrees of freedom: eight at the actual cutoff and two at each placebo. The filled point in the shaded column is the actual cutoff, 2007; hollow points are placebo cutoffs. Placebos at 2002–2004 use only the pre-reform cohorts 2000–2005; placebos at 2009–2011 use only the post-reform cohorts 2007–2012. Outcomes as in Table 10. Shares in percentage points; wage in log points × 100.

Table 10: Effect of the reform at the actual and at placebo cutoffs

	(1)	(2)	(3)	(4)	(5)	(6)
			Degree, by what the reform did to the field			
Cutoff cohort	Pr(university)	Years of univ.	5 to 3+2	4+2 to 3+2	No change	Log wage
2007 (actual)	0.024**	0.229***	0.023***	0.016***	−0.021***	0.021***
	(0.009)	(0.058)	(0.004)	(0.003)	(0.004)	(0.005)
2002 (placebo)	−0.014	0.000	−0.001	−0.006	−0.004*	−0.006
	(0.005)	(0.053)	(0.004)	(0.004)	(0.001)	(0.005)
2003 (placebo)	−0.010	0.017	0.002	−0.003	−0.007*	−0.002
	(0.007)	(0.012)	(0.003)	(0.003)	(0.002)	(0.003)
2004 (placebo)	−0.002	0.078**	0.004*	0.000	−0.005	0.001
	(0.009)	(0.012)	(0.001)	(0.003)	(0.005)	(0.002)
2009 (placebo)	0.004	0.000	0.001	−0.001	0.006*	0.003
	(0.004)	(0.026)	(0.005)	(0.005)	(0.002)	(0.006)
2010 (placebo)	−0.013**	−0.070***	−0.010**	0.001	−0.001	−0.004**
	(0.002)	(0.006)	(0.002)	(0.002)	(0.003)	(0.001)
2011 (placebo)	−0.015	−0.092	−0.008	−0.002	−0.003	−0.010*
	(0.009)	(0.044)	(0.006)	(0.001)	(0.003)	(0.003)
Largest placebo / actual	0.63	0.40	0.42	0.38	0.34	0.46
Largest same sign / actual	0.17	0.34	0.18	0.06	0.34	0.13

Notes: *Quadros de Pessoal*, one observation per worker. Estimates of τ in equation (2) at the cutoff cohort in each row, from a line on each side of the cutoff fitted to the cohort means. The 2007 row is the main estimate (cohorts 2000–2012, 2006 excluded). "Largest placebo / actual": the largest of the six placebo estimates in absolute value, divided by the absolute value of the actual jump. "Largest same sign / actual": the same among the placebo estimates with the sign of the actual jump. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

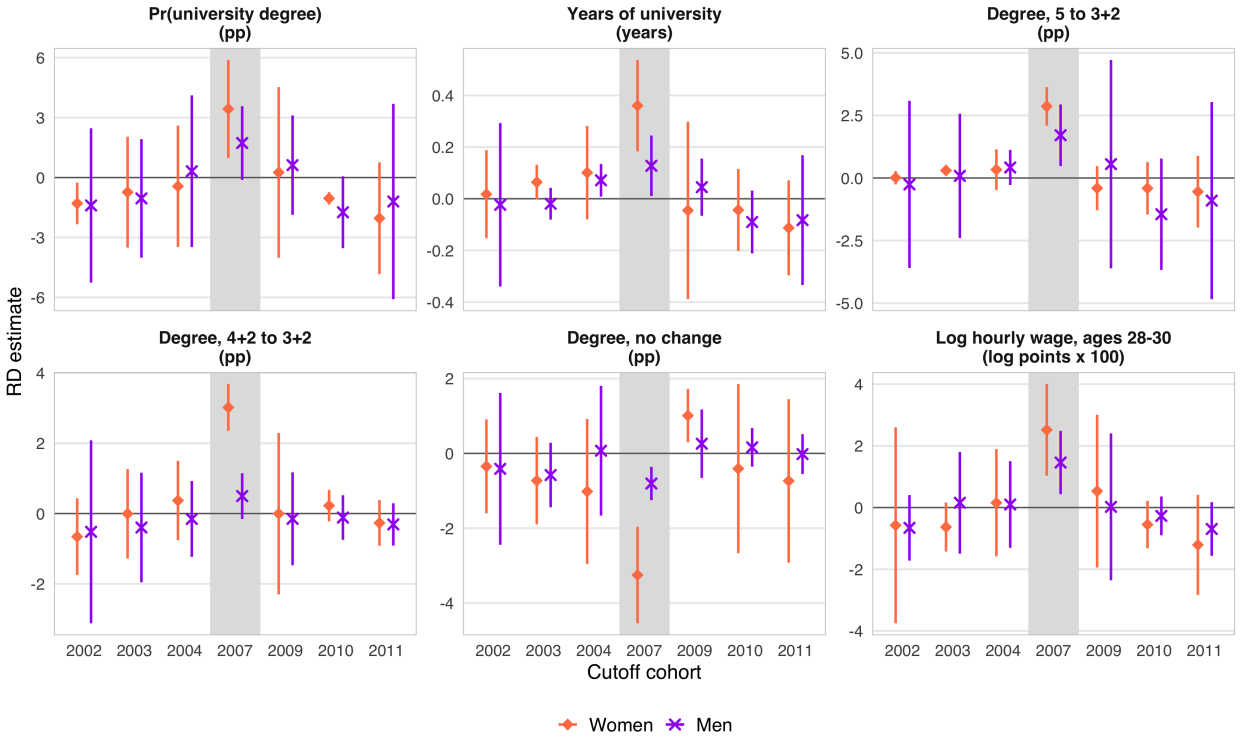


Figure 25: Jumps at the actual and at placebo cutoffs, women and men

Notes: As Figure 24, for women and men separately. The shaded column is the actual cutoff.

Table 11: Effect of the reform at the actual and at placebo cutoffs, women and men

	(1)	(2)	(3)	(4)	(5)	(6)
			Degree, by what the reform did to the field			
Cutoff cohort	Pr(university)	Years of univ.	5 to 3+2	4+2 to 3+2	No change	Log wage
Panel A. Women						
2007 (actual)	0.034**	0.360***	0.029***	0.030***	−0.032***	0.025***
	(0.011)	(0.077)	(0.003)	(0.003)	(0.006)	(0.006)
2002 (placebo)	−0.013**	0.017	0.000	−0.007	−0.003	−0.006
	(0.002)	(0.040)	(0.001)	(0.003)	(0.003)	(0.007)
2003 (placebo)	−0.007	0.064*	0.003***	0.000	−0.007	−0.006*
	(0.006)	(0.016)	(0.000)	(0.003)	(0.003)	(0.002)
2004 (placebo)	−0.004	0.101	0.003	0.004	−0.010	0.002
	(0.007)	(0.042)	(0.002)	(0.003)	(0.005)	(0.004)
2009 (placebo)	0.003	−0.045	−0.004	0.000	0.010**	0.005
	(0.010)	(0.080)	(0.002)	(0.005)	(0.002)	(0.006)
2010 (placebo)	−0.010***	−0.044	−0.004	0.002	−0.004	−0.006*
	(0.001)	(0.037)	(0.002)	(0.001)	(0.005)	(0.002)
2011 (placebo)	−0.020*	−0.113	−0.005	−0.003	−0.007	−0.012*
	(0.007)	(0.043)	(0.003)	(0.002)	(0.005)	(0.004)
Largest placebo / actual	0.59	0.31	0.19	0.22	0.31	0.48
Largest same sign / actual	0.07	0.28	0.12	0.12	0.31	0.21
Panel B. Men						
2007 (actual)	0.017*	0.127**	0.017**	0.005	−0.008***	0.015**
	(0.008)	(0.051)	(0.005)	(0.003)	(0.002)	(0.004)
2002 (placebo)	−0.014	−0.023	−0.003	−0.005	−0.004	−0.007
	(0.009)	(0.073)	(0.008)	(0.006)	(0.005)	(0.002)
2003 (placebo)	−0.010	−0.019	0.001	−0.004	−0.006	0.002
	(0.007)	(0.014)	(0.006)	(0.004)	(0.002)	(0.004)
2004 (placebo)	0.003	0.071**	0.004	−0.002	0.001	0.001
	(0.009)	(0.015)	(0.002)	(0.003)	(0.004)	(0.003)
2009 (placebo)	0.006	0.045	0.006	−0.001	0.003	0.000
	(0.006)	(0.026)	(0.010)	(0.003)	(0.002)	(0.006)
2010 (placebo)	−0.017*	−0.090*	−0.014	−0.001	0.002	−0.003
	(0.004)	(0.028)	(0.005)	(0.001)	(0.001)	(0.001)
2011 (placebo)	−0.012	−0.083	−0.009	−0.003	0.000	−0.007*
	(0.011)	(0.058)	(0.009)	(0.001)	(0.001)	(0.002)
Largest placebo / actual	1.01	0.71	0.85	1.05	0.72	0.48
Largest same sign / actual	0.36	0.56	0.32	0.00	0.72	0.11

Notes: As Table 10, for women and men separately.

H Degrees by field of study

Figure 26 splits the field groups of Section 5.2 into single fields, with STEM split into engineering and computing and into the sciences and mathematics, for women, men and everyone together. The last row is the share of the cohort with a degree whose field is unknown.

I Results in other data sources

This appendix reports the main education results in two Portuguese data sources: *Quadros de Pessoal* and the Labour Force Survey. Table 12 covers university completion and years of university (Section 5.1);

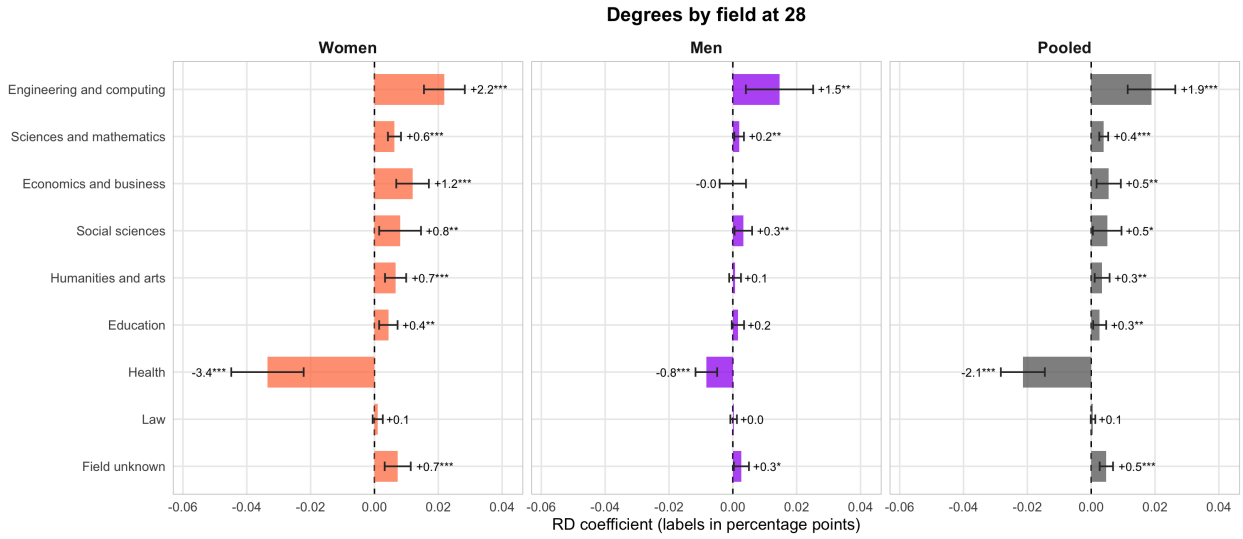


Figure 26: Effect of the reform on university degrees, by field of study

Notes: *Quadros de Pessoal*, workers observed at both 28 and 29, as in Figure 5, one observation per worker, cohorts 2000–2012 (2006 excluded). Each bar is a separate estimate of τ in equation (2) for the share of the cohort holding a university degree in the field on the worker's age-28 record, with a 95 percent confidence interval; labels in percentage points. Fields are ISCED 1997 groups. Engineering and computing: computing, engineering, manufacturing and processing, and architecture and construction (ISCED 48, 52, 54, 58). Sciences and mathematics: life and physical sciences, mathematics and statistics (ISCED 42, 44, 46). Both use the field as reported. Economics and business: business and administration (ISCED 34). Health: health (ISCED 72), together with agriculture and veterinary science (ISCED 62 and 64), mostly nursing. For the other rows a missing field is filled in from the occupation, as in Figure 5; "field unknown" is what remains.

Table 13 covers degrees by what the reform did to the field (Section 5.2).

Table 12: Effect of the reform on university completion and years of university, two data sources

	Highest education is a university degree		Years of university	
	QP	LFS	QP	LFS
All	0.024**	0.024***	0.229***	0.126***
	(0.009)	(0.003)	(0.058)	(0.015)
Baseline	0.307	0.181	2.158	0.772
Women	0.034**	0.037***	0.360***	0.185***
	(0.011)	(0.004)	(0.077)	(0.024)
Baseline	0.386	0.229	2.652	0.938
Men	0.017*	0.012***	0.127**	0.070***
	(0.008)	(0.003)	(0.051)	(0.020)
Baseline	0.231	0.133	1.683	0.608
p-value, women = men	0.199	0.000	0.011	0.000

Notes: RD estimates at the 2007 cohort, excluding the 2006 cohort, with a separate linear trend in the running variable on each side. QP: the estimates of Figure 3, one observation per worker on the age-28 record, from the twelve cohort means; baseline is the mean over the pre-reform cohorts; women = men: z-test. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Years of university are zero for non-graduates; in QP they are proxied by the age at which a worker is first observed with their highest degree, minus 18 (Appendix C.1).

Table 13: Effect of the reform on university degrees by what the reform did to the field, two data sources

	5 to 3+2		4+2 to 3+2		No change	
	QP	LFS	QP	LFS	QP	LFS
All	0.023***	0.006***	0.016***	0.022***	−0.021***	−0.007***
	(0.004)	(0.002)	(0.003)	(0.002)	(0.004)	(0.001)
Baseline	0.086	0.038	0.107	0.081	0.067	0.037
Women	0.029***	0.008***	0.030***	0.037***	−0.032***	−0.010***
	(0.003)	(0.002)	(0.003)	(0.004)	(0.006)	(0.003)
Baseline	0.059	0.027	0.146	0.111	0.107	0.062
Men	0.017**	0.002	0.005	0.009***	−0.008***	−0.003**
	(0.005)	(0.002)	(0.003)	(0.002)	(0.002)	(0.001)
Baseline	0.112	0.049	0.069	0.051	0.028	0.013
p-value, women = men	0.067	0.077	0.000	0.000	0.000	0.012

Notes: RD estimates at the 2007 cohort, excluding the 2006 cohort, with a separate linear trend in the running variable on each side. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The outcome is the share of the cohort whose highest completed level is a university degree in each group of fields. QP: the estimates in Figures 5 and 19.

J Field choices in ministry records

This appendix checks the shift in records kept by the Ministry of Higher Education, which observe students when they apply, when they enter and when they graduate. These records cover public and private institutions and every student, whatever they later do. They are counts by calendar year, so they do not allow the regression discontinuity design of Section 4. I compare the years before and after the reform and read the results as descriptive evidence. Because the age at entry did not change with the reform, the year of entry corresponds closely to the cohort in the main analysis.

Data. I use three records. The first is the national contest for public institutions, run by the Directorate-General for Higher Education (DGES), for 2004 to 2009. For each programme it reports the applications received and the students placed in the first round, both by sex. An applicant can list up to six programmes, and every listed option counts as an application, so applications measure demand while placements are limited by the number of places. The second record counts all students enrolled in the first year of a first degree for the first time, in public and private institutions, for 2004/05 to 2010/11. The third counts the first degrees awarded each year from 2006/07 to 2014/15. Both come from the ministry's statistics office (GPEARI, later DGEEC) and report each programme by sex. I group programmes into three fields: engineering covers engineering, manufacturing and processing, and architecture and construction; economics and business covers economics and business and administration; nursing covers the programmes named *Enfermagem*. For each source and sex I compute the share of all applications, placements or first-year students that falls in each

group. Before the reform means 2004 and 2005, and after means 2007 to 2009. I leave out 2006, as in the main analysis, because 43 percent of programmes had already converted to the new structure that year (Section 2).

Table 14 shows the same shift as *Quadros de Pessoal* in every record. After the reform, a larger share of women applied to and enrolled in engineering. A larger share of applications from both women and men went to economics and business. A much smaller share of women and men applied to and enrolled in nursing.

Table 14: Share of applications, placements and first-year students by field, before and after the reform

		Women			Men		
		Before	After	Change	Before	After	Change
Engineering	Applications (DGES)	8.7	10.9	+2.2	24.6	24.9	+0.4
	First-year students	7.6	9.6	+2.0	27.6	32.2	+4.6
Economics and business	Applications (DGES)	12.5	15.0	+2.5	14.2	17.2	+3.0
	First-year students	15.7	18.7	+3.0	18.5	19.7	+1.2
Nursing	Applications (DGES)	10.9	6.6	−4.3	3.8	1.9	−1.8
	First-year students	8.8	6.5	−2.3	2.7	1.7	−1.0

Notes: Percent of all applications or first-year students of the same sex. Before: average of 2004 and 2005. After: average of 2007, 2008 and 2009. Change in percentage points, computed before rounding.

K Graduate Oversupply

The reform raised the number of new graduates in engineering and in economics and business. If the labour market could not absorb them, post-reform graduates would start in worse jobs or in jobs outside their field. This appendix provides suggestive evidence on this. Conditioning on graduates in each field, I check if there is an increase in the share of graduates starting as trainees, and I check the match-rate, i.e. the share of graduates working in the same occupation or field as the one they studied.

Sample. Unlike the main results, this appendix follows graduates from their first job, so its outcomes are not measured at a fixed age. I use graduate entrants: workers in their first year in QP with a university degree, hired in the last twelve months, aged 21–28. Cohorts are the worker's modal cohort, and the 2006 cohort is excluded. I follow each entrant's main job in the year of the first job and in each of the next three years. In year k the sample keeps entrants whose first job came by age $29 - k$, so every cohort is observed k years later. All estimates are cohort RDs, as in the rest of the paper.

K.1 More trainees, but only for a year

QP records each worker's level in the firm, and the lowest level is reserved for trainees and apprentices. Before the reform, 12.0 percent of female and 11.5 percent of male graduate entrants started on this grade. Table 15 shows that this share jumps at the cutoff, by 3.1 percentage points for women and 2.5 for men. The jump, however, does not last. One year after the first job, the jump is 0.4 percentage points for women and not significant. For men it is −0.9 points. Two and three years after the first job, the jump is zero or negative for both.

Table 15: Trainee grade by years since the first job, all graduate entrants (percentage points)

	First job	1 year later	2 years later	3 years later
Women				
RD jump	3.1***	0.4	-0.2	-0.5
	(0.9)	(0.3)	(0.3)	(0.3)
Pre-reform baseline	12.0	5.2	3.8	2.7
Men				
RD jump	2.5**	-0.9***	-1.1***	-0.9**
	(1.0)	(0.3)	(0.3)	(0.3)
Pre-reform baseline	11.5	4.3	3.0	2.1
Graduate entrants	316,640	316,640	284,324	245,563

Notes: *Quadros de Pessoal*, graduate entrants of all fields (see text). Outcome: main job on the trainee grade (level 8) in the year of the first job and one to three years later; zero for an entrant in QP on another grade. Cohort RD estimates with standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

K.2 No change in field match where graduates rose

The match rate is the share of a field's graduate entrants whose main job is in the sector, or the occupation, that matches the field. Engineering graduates match IT and engineering firms, and engineering and ICT professionals and technicians. Economics and business graduates match finance and business services, and business and finance professionals. Table 16 reports the jump in the match rate at the first job and in the next three years. In engineering, no estimate is significant, for either sex or either match. All lie within about 3 percentage points of zero. In economics and business, the occupation match does not fall. The sector match is 3 to 6 percentage points lower, on a baseline of 34 to 37 percent, but only two of the eight estimates are significant, both at the 10 percent level.

Overall, this section provides suggestive evidence that the over-supply of graduates was absorbed by the labour market.

Table 16: Field match rate by years since the first job, by field (percentage points)

Field	Match by	First job	1 year later	2 years later	3 years later	Baseline
Women						
Engineering	Sector	-0.7	2.0	0.5	0.9	24.2
		(1.2)	(1.8)	(0.9)	(2.0)	
	Occupation	-0.3	-0.4	1.3	2.6	43.3
		(1.7)	(2.4)	(1.5)	(3.0)	
Economics and business	Sector	-2.6	-2.7	-3.9*	-2.5	33.7
		(1.4)	(1.8)	(1.9)	(1.7)	
	Occupation	1.3	0.5	-0.7	-2.2	22.5
		(1.5)	(1.4)	(1.1)	(1.5)	
Men						
Engineering	Sector	-2.3	-2.7	-3.1	-2.6	29.3
		(2.2)	(2.4)	(2.5)	(2.7)	
	Occupation	0.4	-0.4	-1.6	-1.7	65.3
		(0.9)	(1.2)	(1.9)	(1.2)	
Economics and business	Sector	-3.6	-6.2*	-5.6	-3.2	37.0
		(2.2)	(3.1)	(3.5)	(3.3)	
	Occupation	2.4	2.7*	0.6	1.9	23.4
		(1.8)	(1.4)	(2.1)	(3.2)	

Notes: Quadros de Pessoal, graduate entrants of each field who are in QP that year (see text). Outcome: main job in the sector or the occupation that matches the field. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

L The model: details

L.1 Estimation

Exact shares. The estimation evaluates the model many times, and random draws would make the distance jump from one evaluation to the next. I therefore compute the model's shares exactly, by adding up the chance of each path on a grid of 20,000 starting beliefs.

What is fixed. Table 17 lists the numbers I fix and where they come from. Starting beliefs come from PISA 2003 for Portugal, whose students reached university age just before the reform. For each student I predict the chance that she expects to finish university from her mathematics, reading and science scores and from four measures of her confidence in mathematics: anxiety, self-concept, self-efficacy and motivation. I summarise these predictions with the Beta distribution that has the same mean and variance, Beta(0.94, 0.91). Its mean is 0.51. Its 10th and 90th percentiles, 0.09 and 0.91, match those of the predictions, and its median is 0.51 against 0.53.

Table 17: Numbers fixed outside the estimation

		Value	Source
Discount factor	β	0.95	Stange (2012)
Dislike of risk	γ	2	Johnson (2013)
Working life	T	40 years	set by hand
Direct cost of a year of study	c	$0.1\,w_0$	tuition cap, EUR 901–949 a year
Pay, IC five-year degree	w_5^{IC}	1.6	QP
Pay, S four-year first degree	w_4^S	1.4	QP
Pay, S full degree	w_F^S	1.5	QP
Pay, N degree	w^N	1.4	QP
Starting belief	p_0	Beta(0.94, 0.91)	PISA 2003

Notes: Pay in units of the pay of a worker without a degree, $w_0 = 1$. QP: monthly pay at ages 28 to 30 in *Quadros de Pessoal*, relative to secondary-school leavers of the same age and year. Tuition cap: the legal maximum fee in public universities in 2005/06 to 2007/08, close to a tenth of the yearly pay of a worker without a degree.

The distance. The estimation has two parts. Given q, σ and the pay, the three average tastes are set so that the model matches the share of the cohort holding a degree in each field before the reform exactly. The remaining numbers minimise the distance

$$D = \sum_m \left(\frac{\hat{x}_m - x_m}{\mathrm{SE}_m} \right)^2 \tag{19}$$

between the model's moments $\hat{x}_m$ and the data x_m, over the two degree jumps, the dropout level and jump, and the five-year jump. The five-year jump counts only when the model falls below it.

Average tastes. Given q, σ and the pay, the chance that an entrant with starting belief p_0 ends with a degree in field k does not depend on the tastes. Only the entry shares do. I set the tastes by repeating

$$\mu_k \;\leftarrow\; \mu_k + \sigma \log \frac{x_k^{\mathrm{pre}}}{\hat{x}_k^{\mathrm{pre}}(\mu)}, \tag{20}$$

where x_k^{pre} is the share of the cohort holding a degree in field k before the reform in the data and $\hat{x}_k^{\mathrm{pre}}(\mu)$ the model's. This is the fixed point of Berry (1994). It converges because a field's share responds more to its own taste than to the others'.

The two pay levels. I search over w_3^{IC}, w_3^S/w_4^S, q and σ together by Nelder–Mead, on a grid of 5,000 beliefs, from 16 starting points: pay set by hand ($w_3^{IC} = 1.3$ and $w_3^S/w_4^S = 0.95$), the estimate on an earlier version of the targets, and 14 random points. The search keeps the bachelor's pay below the five-year degree's, the three-year degree's pay above w_0, and q between 0.01 and 0.70. Thirteen starts end within

0.1 of the best distance, and across them σ ranges from 0.313 to 0.328, q from 0.411 to 0.420, w_3^{IC} rounds to 1.399 in all, and w_3^S/w_4^S ranges from 0.894 to 0.901. The other three end at worse points: distance 3.2 ($\sigma = 0.38$, $q = 0.33$, $w_3^{IC} = 1.30$), 5.8 ($\sigma = 1.06$, $q = 0.13$, $w_3^{IC} = 1.24$) and 60.6 (q at its bound of 0.70).

Fail rate and taste spread. At the two pay levels, I re-estimate q and σ on a grid. The coarse grid has 30 values of q from 0.02 to 0.60 and 30 values of σ from 0.05 to 20, evenly spaced in logs. The fine grid has 41×41 points around the best coarse point. The minimum is at $q = 0.412$ and $\sigma = 0.321$, with a distance of 2.49, inside both grids. For the standard errors, I draw each target 500 times from a normal distribution with its estimate and standard error, keeping the shares before the reform fixed, and refit on the fine grid. The standard deviation of the refitted values is 0.022 for q and 0.104 for $\log \sigma$, or 0.033 for σ. In 13 percent of the draws the refit lands on the edge of the fine grid, so these standard errors are, if anything, too small.

What the tastes mean. The taste spread is 0.32 years of no-degree pay. If a small field became worth a tenth of a year of pay more to every student, its share would rise by about a third ($e^{0.1/0.32} = 1.36$). All three average tastes are negative: on average, students like each field less than work, by 0.6 to 1.7 years of pay, consistent with studying being costly. Without these tastes, many more students would enter university than held a degree before the reform.

The pay levels at the estimates. At a bachelor's pay of 1.400, 87.5 percent of five-year pay, students who are still unsure after three years stop there in large numbers and the five-year jump falls to zero; at 1.405 nobody finishes five years. At the estimate, the bachelor's is worth about as much as it can be while still raising five-year completion. The pay of the three-year first degree is well below the point, about 0.93, above which graduates of S stop at the first degree instead of continuing to the full degree, so almost all of them continue.

Fit and pay. Figure 27 shows the model's effects next to the data, with confidence intervals. Of the model's wage effect of 1.82 log points (Section 6.2), IC contributes 0.88 points, S 1.33 and N −0.40.

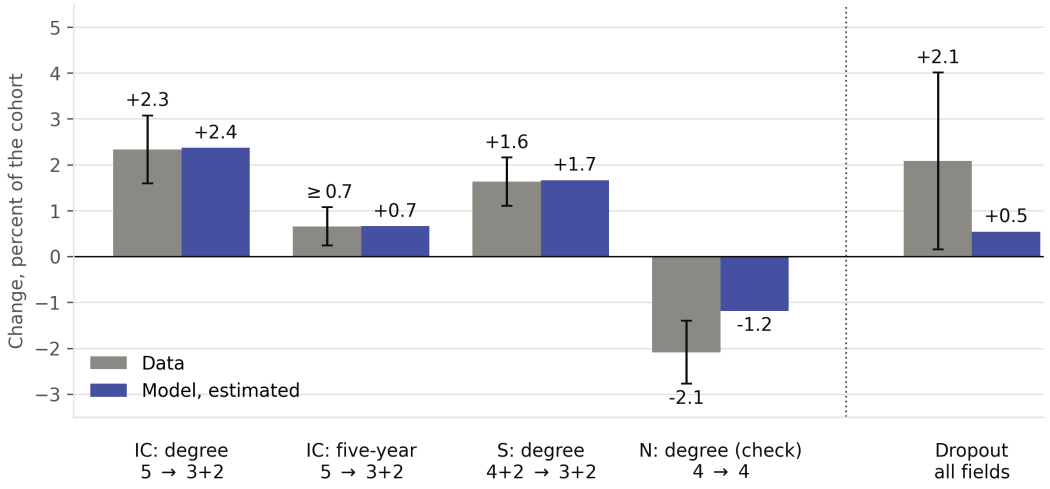


Figure 27: The estimated model and the data

Notes: Effects of the reform in percentage points of the cohort, women and men together, with 95 percent confidence intervals for the data. Degrees by field group: Figure 5. Five-year: work in engineering occupations that require the five-year degree, ages 28 to 30 (Figure 7), a lower bound. Dropout: IEFA (Figure 3), after the dotted line. The jump in N is not targeted. Source: `code/model/toy/plot_paper_counterfactuals.py`. Section 6.2.

L.2 The model without learning

In the model without learning, every student fails each year with one chance $r(p_0)$, the same whether she is a good or a bad match. Passing then tells her nothing, and her belief stays at p_0. I set $r(p_0)$ so that her chance of passing five years is the one she has with learning:

$$\left[1 - r(p_0)\right]^5 = \Pr(\text{pass years 1 to 5} \mid p_0) = p_0 + (1 - p_0)(1 - q)^5. \tag{21}$$

With learning, her chance of failing year $t + 1$, having passed t years, is $(1 - p_t)\, q$, which falls with every year she passes. Without learning it is $r(p_0)$ in every year, and it replaces $(1 - p_t)\, q$ in equation (8). The same $r(p_0)$ applies in every field, so a student passes the six years of S before the reform with probability $[1 - r(p_0)]^6$ and the four years of N with probability $[1 - r(p_0)]^4$. Pay, cost, γ and σ are unchanged. Because the shares before the reform change, I set the average tastes again so that the model matches them, by the inversion of equation (20).

I re-estimate the model without learning by the same search as the estimated model: q, σ, w_3^{IC} and w_3^S / w_4^S, from the same 16 starting points on 5,000 beliefs, and the best point again on 20,000. Without learning, q no longer sets when students fail, only how the chance of finishing differs across students, so I allow it up to 0.99, above the bound of 0.70 used for the estimated model; it is poorly pinned, and starts that

end within 1 of the best distance range from 0.21 to 0.88. With the bachelor's pay free, the best point has $q = 0.21$, $\sigma = 0.62$ and $w_3^{IC} = 1.158$, with a distance of 6.01. Even then dropout falls by 0.3 points instead of rising, because the credential now catches more of the students who fail; this one moment accounts for almost all of its distance. With the bachelor's pay held at 1.387, 86.8 percent of five-year pay as in *Quadros de Pessoal*, the best point has $q = 0.085$, $\sigma = 1.40$ and average tastes of -4.3 to -3.2 years of pay, with a distance of 17.4. Bad matches then fail only 9 percent of years, so that dropout before the reform falls to 3.9 percent. It fails the check that fewer than one entrant in ten would enter even if sure to leave after one year: 16 percent in IC, 30 in S and 29 in N. Holding q at its estimate, 0.412, instead does not help: the distance is then 181, because degrees in IC rise by 7.1 points. The model with learning, re-estimated in the same way, returns the estimates when the bachelor's pay is free (distance 2.49), and has $q = 0.397$, $\sigma = 0.331$ and a distance of 2.67 when the bachelor's pay is held at the QP ratio.

L.3 When the intermediate credential comes

Table 18 reports the effects of the reform with the intermediate credential after two, three or four years, the counterfactual of Section 6.3.

Table 18: When the intermediate credential comes

	Credential after		
	2 years	3 years	4 years
Edge, % of five-year pay	82.3	87.8	93.7
Bad matches who fail after the credential, %	27.5	13.3	4.9
A. Paid as close to its edge as the estimated bachelor's			
Pay, % of five-year pay	82.0	87.4	93.3
Δ degree in IC	+7.59	+2.37	+0.37
Δ five-year degree	−2.21	+0.66	+0.30
Δ dropout	+1.14	+0.54	+0.36
B. Paid what raises five-year degrees the most			
Pay, % of five-year pay	80.0	86.5	93.2
Δ degree in IC	+4.94	+2.16	+0.37
Δ five-year degree	+3.88	+1.88	+0.30
Δ dropout	+0.58	+0.48	+0.36

Notes: Effects of the reform in percentage points of the cohort, with the credential in IC after two, three or four years, the shorter first degree in S as estimated, and everything else at the estimates. Degree in IC: any credential, including the intermediate one. Edge: the pay at which a student sure of a good match stops at the credential. Bad matches who fail after the credential: share of bad matches who enter and fail after passing that many years. Panel A: 99.6 percent of each edge, the estimated bachelor's share of its own; the column for three years is the estimated model.

Below its edge, the credential after two years raises five-year completion the most: by up to 3.9 percentage points of the cohort, at 80 percent of five-year pay, against 1.9 for the bachelor's after three years and 0.3 for a credential after four. It also draws in the most students, and dropout rises by about as much as with the bachelor's. But it must pay less. Paid as close to its edge as the estimated bachelor's, at 82 percent of five-year pay, 10 percent of the cohort leave with it after two years, and five-year degrees fall by 2.2 points. If it paid the same premium per year of study as the estimated bachelor's, it would pay 79 percent of five-year pay, close to its peak.

L.4 Who enters each field and who finishes five years

Figure 28 shows who chooses each option before and after the reform, by starting belief. The reform draws 3.4 percent of the cohort out of work. These students are neither the least nor the most sure of their match: four in five start with a belief between 0.46 and 0.83, and their average starting belief is 0.65, against 0.37 among students who work before the reform. They split almost evenly between IC and S, 1.7 percentage points of the cohort each, and none enter N. Entry into IC rises from 9.6 to 12.5 percent. The students the credential draws in are less sure of their match than those already there: their average starting belief is 0.72, against 0.89 among students who entered IC before the reform. Besides students who would have worked, they include students who would have entered N, 0.8 points of the cohort, and S, 0.5 points. Entry into S rises from 13.1 to 15.0 percent, and its new students are also less sure than those already there, at 0.70 against 0.79. N loses 1.4 points of the cohort to the two reformed fields, from 8.2 to 6.8 percent, and the students it loses start about as sure as those who stay, at 0.81 against 0.79.

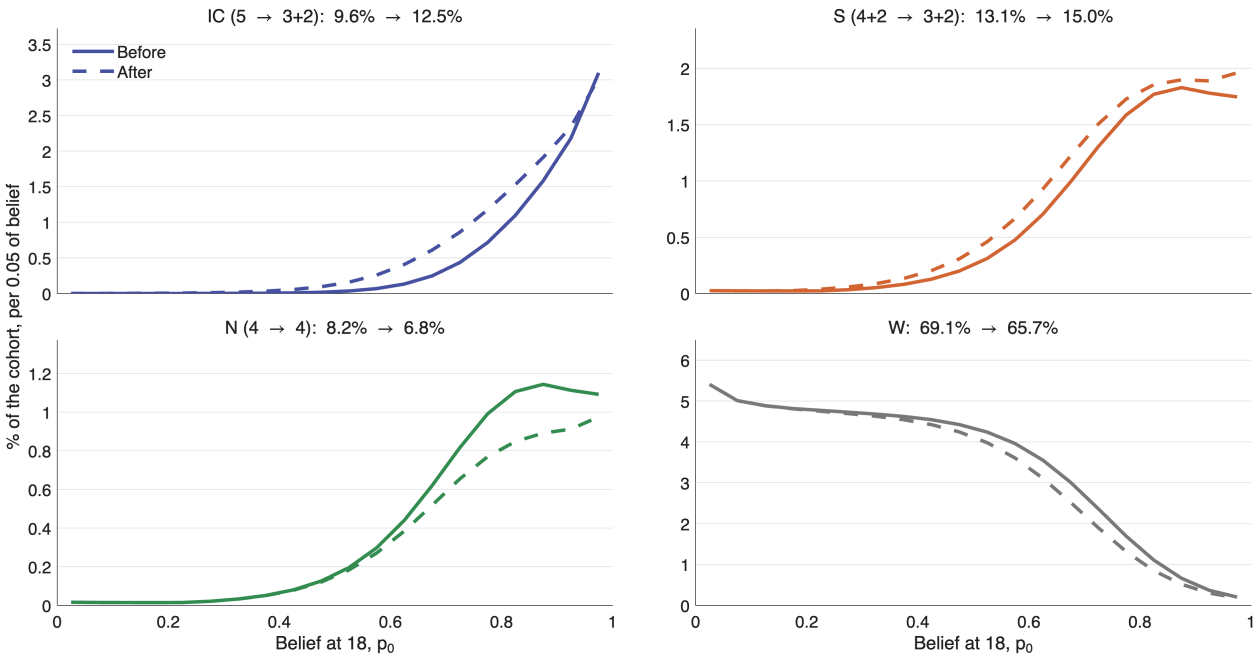


Figure 28: Who enters each option, by starting belief, in the estimated model

Notes: Share of the cohort choosing each option per 0.05 of starting belief, before (solid) and after (dashed) the reform; panel titles give the total share of the cohort. Starting beliefs from PISA 2003 (Table 17).

The credential also changes who finishes five years in IC. Before the reform, a student who had passed three years had nothing to stop with, so she continued. After it, she stops with the bachelor's unless passing has raised her belief to at least 0.93. This bar comes from the estimated pay of the bachelor's; a lower pay would lower it. After three passed years, it means a starting belief of at least 0.73, and 28 percent of the cohort start above it. The bar splits the students the credential draws into IC about in half. Those above it go on to five years unless they fail, and most of them finish. Those below it stop with the bachelor's if they pass three years, and leave with nothing if they fail before.

This bar sorts the flows into and out of the five-year degree reported in Section 6.3. In the estimated model, 1.3 percent of the cohort finish five years who would not have done so before the reform. All of them start above the bar, with an average starting belief of 0.83, and almost all of them are good matches. None of them would have entered IC before the reform: 0.5 points of the cohort would have worked, 0.5 points would have graduated in N, and 0.3 points would have taken the four-year first degree in S. Most of them, then, would have graduated anyway, in another field. Going the other way, 0.5 percent of the cohort stop at three who would have finished five years. They enter IC before and after the reform, and they start just

below the bar: four in five start between 0.57 and 0.73, with an average of 0.66. Almost all of them, 96 percent, are good matches. After three passed years their belief averages 0.90, and at that belief two more years, with the risk of failing in them, are not worth the 14 percent more that the five-year degree pays. A further 0.1 percent of the cohort, among the most confident students, would have finished five years in IC and now take the full degree in S instead, which became a year shorter. On net, five-year degrees rise by 0.66 percentage points.

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